FINITE QUOTIENT TOWERS AND GEOMETRIC WITT FILTRATIONS

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ABSTRACT. We describe Rezk's Witt filtration $\{I_d\}$ through the geometry of finite-subgroup quotients. For a one-dimensional formal group Γ/k of finite height h over a perfect field of characteristic p, we write $\mathcal{O} = W(k)[[u_1, \dots, u_{h-1}]]$ for its universal deformation ring. A fixed mixed-characteristic source lift determines the algebra B_d of order-p^d quotient targets and the endpoint ideal $E_d = \ker(\mathcal{O} \to B_d)$. We prove the integral recursion $I_{d+1} = pI_d + E_d$. The primitive Euler image $\bar{c}_d \in B_d$ gives the stronger presentations $\mathcal{O}/I_d \simeq B_d/(\bar{c}_d)$ and $\mathcal{O}/I_{d+1} \simeq B_d/(p\bar{c}_d)$. Comparing the resulting minimal resolutions shows that every endpoint ideal is a complete intersection of codimension $h-1$ and every nontrivial Witt ideal is a complete intersection of codimension h. We also determine the Smith factors of the ordinary trace pairing, construct a perfect trace linking form on each Witt truncation, and characterize the shallower Witt ideals by trace divisibility in a single endpoint algebra. At height two, we obtain $I_r = (H_{r-1}, H_r)$, where H_d is the fixed-source endpoint equation. A Hecke calculation in a preferred coordinate proves this formula for the reference source; source change and descent extend it to every chosen source lift. Thus, consecutive endpoint divisors recover the Witt filtration scheme-theoretically. Determinant norms relate these constructions to formal resultants, Lubin–Tate–Witt ghost coordinates, and affine branch towers.

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1. Introduction

We study Rezk's Witt filtration on a Lubin–Tate deformation ring. We describe its ideals in terms of the targets of finite-subgroup quotients of a fixed mixed-characteristic source lift. This description gives an integral

recursion, complete intersection presentations, and trace-theoretic criteria for membership in the filtration.

Finite subgroup quotients occur in arithmetic geometry and stable homotopy theory [24, 22, 7, 28]. Quotients of algebraic groups give finite covers with affine Galois actions on their generic branches. Their formal completions are described by prepared power series and resultants. In Morava E-theory, universal finite subgroups determine power operations and Lubin–Tate–Witt ghost coordinates. We compare these constructions through the finite locally free quotient algebra.

The comparison uses the determinant norm. For a finite locally free map $q : Y \to X$, multiplication by a section $a \in A_q = q_*\mathcal{O}_Y$ is an endomorphism of a finite projective module, and its determinant is the norm $N_q(a)$. If the generic fiber is étale, a faithfully flat splitting identifies this determinant with the product of the values of a on the sheets of the cover. For a monogenic completed algebra, it is a resultant and along a chain of quotient maps it is transitive. These are the standard determinant-norm properties for finite locally free maps [10, 38]. They are enough to compare the ordered branch tower, the formal quotient tower, and the universal power-operation tower without choosing separate transition laws for each of them.

The local problem considered here begins with a one-dimensional formal group Γ of height h over a perfect field k of characteristic p. Its universal deformation ring is

$$\mathcal{O}_h = W(k)[[u_1, \dots, u_{h-1}]].$$

This is the Lubin–Tate deformation space [24, 14]. Strickland's finite subgroup schemes provide finite free algebras over $\mathcal{O}_h$ that classify subgroup schemes of prescribed order [39, 41]. Ando's isogeny description of power operations and Zhu's norm-coherent coordinates identify the quotient coordinate with a finite-free norm [2, 46]. Rezk packages all of the universal quotient data in the Lubin–Tate–Witt functor and equips $\mathcal{O}_h$ with a decreasing filtration

$$\mathcal{O}_h = I_0 \supset I_1 \supset I_2 \supset \cdots$$

that interpolates between the p-adic filtration and the geometry of finite subgroups in characteristic p [33, 29]. The integral geometric meaning of this filtration is the question addressed below.

1.1. **Fixed-source endpoint ideals.** For a subgroup of order p^d, the universal subgroup algebra has two maps from the deformation ring: a source map and a target map [22, 39]. We specialize the source to a chosen mixed-characteristic lift of the formal group and retain the target. Equivalently, choose parameters and a section $\epsilon_W : \mathcal{O}_h \to W(k)$ with $u_i \mapsto 0$. This gives a finite free $W(k)$-algebra B_d and a surjection

$$\tau_d : \mathcal{O}_h \longrightarrow B_d.$$

Its kernel E_d is the fixed-source endpoint ideal. Scheme-theoretically, $V(E_d)$ is the image of all targets obtained by quotienting the fixed source lift by

subgroup schemes of order p^d. At height two this image is a divisor and E_d is principal. At height three it has codimension two. In larger height its minimal resolution has the Koszul ranks established in theorem 8.3. The definition of E_d does not require a choice of generators, and it continues to make sense when no scalar endpoint equation exists.

Our first main theorem identifies Rezk's filtration recursively with these endpoint ideals.

Theorem 1.1 (Geometric Witt recursion)**.** *For every finite height h and every $d \geq 0$,*

$$I_{d+1} = pI_d + E_d. \tag{1.1}$$

The formula determines the integral carry. Retaining the full Euler–trace map yields the stronger presentation below.

Theorem 1.2 (Euler presentations and complete intersections)**.** *Let $\bar{c}_d \in B_d$ be the primitive Euler image. Then $\mathcal{O}_h/I_d \simeq B_d/(\bar{c}_d)$ and $\mathcal{O}_h/I_{d+1} \simeq B_d/(p\bar{c}_d)$, with the target coefficient convention. Every E_d is generated by a regular sequence of length $h-1$, and every I_r, $r \geq 1$, by a regular sequence of length h. The minimal free resolutions over $\mathcal{O}_h$ of the quotient rings $\mathcal{O}_h/E_d \simeq B_d$ and $\mathcal{O}_h/I_r$ have Betti polynomials $(1+t)^{h-1}$ and $(1+t)^h$, respectively.*

Two consecutive Witt quotients arise from the same endpoint algebra by killing $\bar{c}_d$ and $p\bar{c}_d$. Their minimal mapping cones have equal Betti numbers, so the Koszul ranks propagate from I_1 to every depth.

The quotients have a uniform p-adic structure. Put

$$N_{h,d} = \#\{D \leq (\mathbb{Q}_p/\mathbb{Z}_p)^h : |D| = p^d\} = \begin{bmatrix} d+h-1 \\ h-1 \end{bmatrix}_p.$$

The Gaussian coefficient is the classical subgroup-counting polynomial [3, 25]. The successive p-adic layers of $\mathcal{O}_h/I_r$ have dimensions $N_{h,r-s-1}$, and the complete invariant-factor decomposition is as follows.

Theorem 1.3 (Mixed-characteristic layers)**.** *For $r \geq 1$ there is a noncanonical isomorphism of $W(k)$-modules,*

$$\mathcal{O}_h/I_r \simeq \bigoplus_{e=1}^{r} (W(k)/p^e)^{\oplus\left(N_{h,r-e}-N_{h,r-e-1}\right)}, \tag{1.2}$$

where $N_{h,-1} = 0$.

This determines every mixed-characteristic layer. Combined with the Euler presentation, it also determines the complete Smith spectrum of the ordinary trace pairing: exponent a occurs $N_{h,a} - N_{h,a-1}$ times for $0 \leq a \leq d$. More intrinsically, for $0 \leq s \leq d$ one has

$$I_s = \{x \in \mathcal{O}_h : \mathrm{Tr}^{\mathrm{source}}_{B_d/W(k)}(\tau_d(x)B_d) \subseteq p^s W(k)\}.$$

Thus a single sufficiently deep endpoint algebra detects every shallower Witt level by ordinary trace divisibility; see theorems 8.6 and 8.7.

1.2. **The height-two intersection theorem.** At height two, the fixed-source endpoint image is a Cartier divisor. Let H_d denote its monic equation in the target coefficient convention. If P_d is the characteristic polynomial formed over the source copy of $W(k)$, then $H_d = \varphi^{-d}(P_d)$ coefficientwise; see lemma 9.4. Rezk's filtration then satisfies the recursive formula $I_{d+1} = pI_d + (H_d)$. The one-dimensional deformation space has an additional Hecke geometry, expressed through modular isogeny complexes and the Bruhat–Tits tree [32, 45, 37], that yields a sharper description. For Zhu's preferred coordinate and its zero source, the exact-distance endpoint polynomials satisfy a uniform congruence modulo p^2 and an all-depth carry. These identities compute the intersection for that source. A source-change lemma and faithfully flat descent then give the intrinsic formula for an arbitrary source lift.

Theorem 1.4 (Height-two intersection theorem)**.** *Let $h = 2$. For every prime p and every $r \geq 1$,*

$$I_r = (H_{r-1}, H_r). \tag{1.3}$$

In Zhu's preferred coordinate with the zero source, and after scalar extension of that coordinated model, the exact-distance endpoint polynomials C_i, C_j with $i < j$ satisfy

$$(C_i, C_j) = (C_i, p), \qquad \mathcal{O}_2/(C_i, C_j) \simeq k[[u_1]]/(u_1^{\deg C_i}). \tag{1.4}$$

Thus the height-two Witt filtration is the scheme-theoretic intersection filtration of consecutive finite-subgroup endpoint divisors, with the full nilpotent structure determined by the pairwise formula.

1.3. **Finite quotient towers.** The geometric theorem is local, but its construction belongs to a broader quotient-tower formalism. Let G be a connected commutative algebraic group, let $\alpha : G \to G$ be a finite separable isogeny, and let a finite group H normalize α. The induced self-map on G/H has a generic preimage tower whose geometric Galois groups are affine groups built from $G[\alpha^n]$. Under a trivial-stabilizer hypothesis, specialization at a rational point loses no information: the quotient-preimage field is the ordinary Kummer division field [34, 18, 21]. The fixed points of Frobenius elements are solutions of a finite-module affine equation, and the full cycle structure can be encoded by an orbit Witt vector.

The determinant norm gives the symmetric counterpart of this ordered branch tower. In the formal completion it produces the prepared quotient equation and all polynomial resultants. In the universal deformation it produces the Lubin–Tate–Witt ghost coordinate. The trace of multiplication gives the first characteristic coefficient, while the determinant gives the last; the intermediate coefficients are traces on exterior powers. The associated characteristic Witt vector has ghost coordinates $\mathrm{Tr}(b^r)$ [1, 16], and its Hankel matrix is the trace-pairing matrix in a power basis. These constructions extract different invariants of finite quotient algebras. The Euler–trace

comparison and Euler presentation in theorems 5.4 and 8.1 relate the trace pairing on the endpoint algebra to Rezk's filtration.

Theorem 1.5 (Three realizations of a quotient tower)**.** *For a finite locally free subgroup quotient, assume generic étaleness when using branch labels; a one-dimensional formal factor, a monogenic prepared quotient algebra, and norm-coherent coordinates when using resultants; and the universal deformation with its norm-coherent orientation when using Lubin–Tate–Witt ghosts. Under these respective hypotheses, the branch product, completed resultant, and quotient-isogeny ghost are realizations of the same determinant norm. Their composition laws are the corresponding base changes of norm transitivity.*

1.4. **Relation with other work.** The deformation-theoretic background begins with the Lubin–Tate moduli space and Lubin's study of finite subgroup isogenies [24, 22, 14]. Drinfeld's level structures provide the parallel moduli language for finite subgroup data [7]. The finite-subgroup geometry used here has several established components. Strickland constructs the rings that represent subgroup schemes of a formal group and identifies the transfer quotient of Morava E-cohomology of symmetric groups with those rings [39, 41, 42]. Ando's isogeny criterion explains how a multiplicative power operation acts on a complex orientation [2]. Zhu proves that the quotient coordinates can be chosen norm-coherently and, in height two, gives an integral all-prime presentation of the order-p correspondence [46, 44]. At height two, modular isogeny complexes and the Hecke action on Morava E-theory give a complementary description of chains of subgroup quotients [32, 45, 19]. Rezk's work supplies the operation algebra, its Koszul structure, the Lubin–Tate–Witt corepresenter, and the Witt filtration [30, 31, 33]. Cofreeness was proved by Burklund–Schlank–Yuan [4, Theorem 3.4]; here we use Rezk's operation-theoretic formulation and proof [29].

We also note the independent, contemporaneous work of Andrea Taccani [43], which obtains the fixed-source geometric recursion and studies integral trace duality for the Witt filtration.

Pinpoint references to [44] use the numbering of arXiv:1508.03358v2; those to [46] use arXiv:1706.03445v1. The bibliography records both these versions and the journal publications.

The new point is the integral interaction of these objects after fixing the source. The trace–Bockstein bound, Euler–trace comparison, and subgroup-flag argument prove (1.1); the exact comparison map then yields the Euler presentations, complete-intersection results, and trace recognition theorem.

There is a parallel history on the arithmetic side. Iterated endomorphisms of algebraic groups lead naturally to Kummer extensions and affine Galois actions [34, 18, 21]. Their local and reduction behavior is part of the broader arithmetic of abelian varieties [36]. The quotient by a finite group, in the sense of finite group-action quotients [38, Tags 04UV and 07S5], introduces a deck action and a possible constant-field quotient. The exact core computation

in section 2.3 separates these two features. What is new in the present organization is that the affine branch tower is connected to the formal and chromatic constructions by the same finite-free norm, rather than by a comparison made after the individual towers have been computed.

The orbit and characteristic Witt vectors used below are classical finite constructions. Their ghosts record fixed points and power traces, respectively [6, 26, 1, 16]; for characteristic polynomials and resultants, see [25, 13].

1.5. **Outline of the proof.** The proof of theorem 1.1 has three parts. The first is a trace estimate. The triangular p-typical ghost polynomials are recalled in [16, 33]. If $x \in E_s$, the s-th ghost coordinate of its total power operation vanishes. Writing the corresponding p-typical ghost as

$$w_n = \sum_{i=0}^{n} p^i a_i^{p^{n-i}},$$

a least-valuation argument shows that $a_0, \ldots, a_s$ are divisible by p. It follows that the image of x in every other source-sliced endpoint algebra lies in a specified trace lattice. Dividing once by p gives an endpoint Bockstein

$$\beta_d : E_d \longrightarrow B_{d-1}$$

whose image is contained in the $(d-1)$st trace lattice.

The second part computes the colength of that lattice. Strickland's primitive Euler class and transfer quotient [41] identify the transfer quotient with the primitive summand in $E^0 B\Sigma_{p^d}$. The $K(h)$-local Frobenius pairing and the ordinary trace pairing differ by the exact factor p^d. This factor is the centralizer order of a transitive $\mathbb{Z}_p^h$-action of cardinality p^d. The determinant of the comparison map is the norm of the primitive Euler class. After generalized-character base change [17, 12, 40], its component at a subgroup D is the product of the coordinates of the nonzero points of D. Grouping these factors by exact order gives valuation

$$L_d = \sum_{j=0}^{d-1} N_{h,j}.$$

The trace lattice therefore has colength L_d, exactly the colength predicted by Rezk's filtration.

The third part is integral recognition. We use Rezk's congruence criterion, Koszul theorem, and cofreeness [30, 31, 29]. The weight-p pushout supplies a divided Frobenius operation, and the degree-one additive operations form a finite free module. The Koszul theorem implies generation in degree one. After choosing word bases and keeping track of the noncentral coefficient action through balanced tensor products, the resulting operation words lift to polynomial generators of the Lubin–Tate–Witt corepresenter. Membership in I_d is equivalent to the vanishing of the residue-valued coordinates of length less than d. A two-step flag of finite subgroups, in the subgroup-moduli formalism of [7, 39, 32], shows scheme-theoretically that every additive

degree-one operation sends E_d to E_{d-1}. The Bockstein controls the divided Frobenius. The one-step recognition criterion then yields $E_d \subseteq I_{d+1}$, and equality follows from the colength calculation.

There is then a fourth, algebraic step. Injectivity of the trace-dual map upgrades $\lambda_d q_d = p^d \Psi_d$ to $\mathcal{T}^d(B_d) = \bar{c}_d B_d$. This identifies two consecutive Witt quotients with $B_d/(\bar{c}_d)$ and $B_d/(p\bar{c}_d)$. Their minimal mapping cones have equal Betti numbers, forcing complete intersections at every depth. Complementing the Witt invariant factors inside the interval $[0, d]$ gives the trace Smith spectrum, and equal colength then recognizes all shallower trace lattices.

At height two, a separate calculation begins with Zhu's all-prime source–target coordinate [44]. Its reduction modulo p^2 gives a first-order formula for the Hecke correspondence. Applying the signed monic endpoint transform along the Bruhat–Tits tree [37, 32] produces the exact-distance endpoint congruences. The resulting simultaneous carry implies p-saturation, the Euclidean carry, and the pairwise intersection law for the preferred source. The recursive theorem identifies that geometric filtration with the Witt filtration. The source-change and descent arguments of section 9.9 then prove (1.3) without the preferred-source restriction.

2. Finite quotient towers

2.1. **Determinant lines and norm transitivity.**

Definition 2.1 (Finite quotient tower and norm functor)**.** *A finite quotient tower over an integral base X_0 is a chain*

$$X_n \xrightarrow{q_n} X_{n-1} \longrightarrow \cdots \longrightarrow X_1 \xrightarrow{q_1} X_0$$

of finite locally free, generically étale morphisms. For a finite locally free map $q : Y \to X$ of rank n, write

$$A_q = q_* \mathcal{O}_Y, \qquad L_q = \det(A_q).$$

For an invertible $\mathcal{O}_Y$-module $\mathcal{M}$, its norm line is

$$\mathcal{N}_q(\mathcal{M}) = \det(q_* \mathcal{M}) \otimes L_q^{-1}. \tag{2.1}$$

A local section $s : \mathcal{O}_Y \to \mathcal{M}$ induces a norm section $\mathcal{N}_q(s)$ by taking the determinant of $q_* s : A_q \to q_* \mathcal{M}$. In the special case $\mathcal{M} = \mathcal{O}_Y$ and $a \in A_q$, this is

$$N_q(a) = \det(m_a : A_q \to A_q).$$

The trace pairing gives a map $A_q \to A_q^\vee$; its determinant is the section

$$\Delta_q \in \Gamma(X, L_q^{-2}).$$

These constructions commute with arbitrary base change. We use the norm of invertible modules in the sense of Ferrand; see [10] *and* [38, Tag 0BD2].

We recall the standard tower identities for norms and discriminants [10, 38].

Theorem 2.2 (Norm transitivity and discriminants)**.** *Let $Y \xrightarrow{p} Z \xrightarrow{q} X$ be finite locally free of ranks r and s, and generically étale. There is a canonical isomorphism*

$$L_{q\circ p} \simeq L_q^{\otimes r} \otimes \mathcal{N}_q(L_p). \tag{2.2}$$

Under this isomorphism,

$$N_{q\circ p} = N_q \circ N_p, \qquad \Delta_{q\circ p} = \Delta_q^{\otimes r} \otimes \mathcal{N}_q(\Delta_p). \tag{2.3}$$

All three identities commute with arbitrary base change.

Proof. The determinant-line isomorphism is fpqc-local on X. After trivializing the finite projective algebra A_q and the finite projective A_q-module $p_*\mathcal{O}_Y$, the composite algebra is represented by a block matrix over $\mathcal{O}_X$. Taking its top exterior power gives (2.2); taking determinants of multiplication gives norm transitivity in (2.3). These local formulas are compatible with change of trivialization and therefore descend.

On the generically étale locus, the discriminant formula is the usual tower identity for finite separable algebras. Both sides of the second identity in (2.3) are regular sections of the line in (2.2) raised to the power -2. They agree on a dense open set and hence agree on the integral base. The construction uses only determinants, pushforward of finite locally free modules, and tensor products, so it commutes with scalar extension. □

Corollary 2.3 (Discriminant–different section)**.** *Suppose the completed algebra of a finite locally free map is monogenic,*

$$B = A[T]/(W(T, Y)),$$

where W is monic of degree n and generically separable in T. Then

$$\mathrm{Disc}_T(W) = (-1)^{n(n-1)/2}\,\mathrm{Norm}_{B/A}(\partial_T W).$$

The generic branch discriminant and the local different are base changes of this determinant section. See [38, Tag 0BW8] *and* [35, Chapter III].

2.2. **Ordered, formal, and Witt realizations.** Let A/R be an abelian scheme over a complete DVR, let $D \subset A$ be a finite locally free connected subgroup scheme, and put $q_D : A \to A/D$. Write $\mathcal{B}_D = (q_D)_*\mathcal{O}_A$. Quotients and formal isogenies of this form are discussed in [22, 36]. Whenever generic branches are used, we add the hypothesis that q_D is generically étale, equivalently that D_K is finite étale over the fraction field K of R. This hypothesis is automatic in mixed characteristic for the connected p-power subgroup schemes considered below, but not for an arbitrary complete DVR in equal characteristic p.

Proposition 2.4 (Generic splitting)**.** *Assume that D_K is finite étale. Choose a finite separable extension K'/K over which D_K is constant. After base change to K' and pullback along q_D, there is an isomorphism*

$$q_D^*\mathcal{B}_{D,K'} \simeq \prod_{t\in D(K')} \mathcal{O}_{A_{K'}}.$$

For every local section g one has

$$q_D^* N_D(g) = \prod_{t \in D(K')} t^* g.$$

The generic branches form a torsor under $D(K')$.

Proof. The generic quotient is a finite étale D_K-torsor. The torsor identity

$$A_K \times_{(A/D)_K} A_K \simeq A_K \times_K D_K$$

splits the pulled-back quotient algebra. After the chosen extension K'/K, the right-hand side is a disjoint union indexed by $D(K')$. Multiplication by g is diagonal on the resulting product, so its determinant is the product of the branch values. Translation by $D(K')$ acts simply transitively on the factors. □

Proposition 2.5 (Formal completion and resultants)**.** *Assume that the completions of A and A/D at their identity sections contain a one-dimensional connected formal factor through which D acts. Choose a source coordinate T and the norm-coherent target coordinate Y. If the completed quotient algebra is monogenic with prepared equation*

$$\widehat{\mathcal{B}}_D \simeq R[[Y]][T]/(W_D(T, Y)),$$

then, for every polynomial probe $g(T)$,

$$N_D(g) = \mathrm{Res}_T(W_D, g), \qquad Y = N_D(T).$$

Proof. Formal completion is a base change of the finite locally free quotient. The determinant of multiplication by g in the displayed monogenic algebra is its resultant with W_D. Our chosen target coordinate is the target-valued norm $Y = N_D(T)$, so its pullback is $q_D^* N_D(T)$. When existence of a simultaneous norm-coherent system is invoked, the hypotheses of theorem 3.1 are understood. □

Proposition 2.6 (Universal Lubin–Tate realization)**.** *Let F be the universal deformation of a one-dimensional formal group and let D_d be the universal subgroup of order p^d over the corresponding Strickland ring. Assume a norm-coherent coordinate is chosen, for example under the hypotheses of theorem 3.1. In that coordinate,*

$$\mathrm{gh}_d(P(T)) = q_{D_d}^* T_d = \prod_{t \in D_d} (T +_F t).$$

Thus the depth-d Lubin–Tate–Witt ghost is the same finite-free norm coordinate.

Proof. The Strickland chart represents the universal subgroup quotient [39]. Ando's isogeny criterion identifies the total power operation on the complex orientation with the associated quotient isogeny [2]. Norm coherence then identifies its target coordinate with the displayed product. □

Theorem 2.7 (Compatibility of the three realizations)**.** *For a chain $D \subset D'$ of finite locally free subgroups, impose the respective hypotheses of propositions 2.4 to 2.6 whenever using that realization. Restriction of generic branch labels, composition of formal resultants, and comultiplication of the quotient-isogeny ghosts are then obtained by base change from the single identity*

$$N_{D'} = N_{D'/D} \circ N_D.$$

Proof. For a finite locally free algebra $A \to B$, the norm of b is $\det_A(m_b)$. On a faithfully flat splitting of its étale generic fiber, multiplication is diagonal and this determinant is the product of the branch values. For a subgroup quotient these are the translated values, by the torsor identity of proposition 2.4.

For a monogenic algebra $B = A[T]/(W)$ with W monic, the same determinant is $\mathrm{Res}_T(W, g)$ on a polynomial probe g. This identity follows from the root formula over the universal coefficient ring and hence holds without assuming separability. Determinants commute with completion and scalar extension. In the chosen formal coordinate this gives $Y = N_D(T)$.

For the universal deformation, the orientation/isogeny theorem and norm coherence in proposition 2.6 identify the quotient-isogeny ghost with this norm coordinate. This step uses those orientation hypotheses, not a statement about arbitrary orientations. Finally the quotient maps factor as $F \to F/D \to F/D'$. Norm transitivity from theorem 2.2 gives $N_{D'} = N_{D'/D} \circ N_D$. Under the three realizations just described, this is respectively the nested branch product, the composition of resultants, and the composition law for the quotient-isogeny ghosts. □

2.3. **Affine monodromy and specialization.** Let K be a perfect field. A quotient-isogeny datum consists of a connected smooth commutative quasi-projective group G/K, a finite group $H \subset \mathrm{Aut}_K(G, 0)$ whose action is generically free, a finite separable isogeny α, an automorphism ϑ of H satisfying $\alpha h = \vartheta(h)\alpha$, and an H-fixed point $Q \in G(K)^H$. Write $\pi : G \to G/H$ for the quotient map. The affine map $\psi(P) = \alpha(P) + Q$ descends to $f : G/H \to G/H$; for the quotient construction see [38, Tags 04UV and 07S5]. Put

$$A_n = G[\alpha^n](\overline{K}), \qquad K_n = \ker(H \to \mathrm{Aut}(A_n)), \qquad H_n = H/K_n.$$

Each K_n is normal in H, being the kernel of a representation. If $h \in K_{n+1}$ and $T \in A_n$, choose $S \in A_{n+1}$ with $\alpha S = T$. Then

$$\vartheta(h)T = \vartheta(h)\alpha S = \alpha h S = \alpha S = T.$$

Hence $\vartheta(K_{n+1}) \subseteq K_n$, and the transition $H_{n+1} \to H_n$, $hK_{n+1} \mapsto \vartheta(h)K_n$, is well defined.

Theorem 2.8 (Exact affine core quotient)**.** *Over $\overline{K}$, the Galois closure of f^n has group*

$$G_n^{\mathrm{geom}} \simeq A_n \rtimes H_n$$

in its faithful affine action on A_n. The transition map is

$$(S, hK_{n+1}) \longmapsto (\alpha S, \vartheta(h)K_n).$$

Proof. The dominating cover $g_n = \pi \circ \psi^n : G \to G/H$ has deck group $A_n \rtimes H$, acting by $P \mapsto hP + S$, and its order equals the degree of the cover. The source field of f^n is the H-fixed intermediate field. Inside $A_n \rtimes H$,

$$(S, 1)(0, h)(-S, 1) = ((1 - h)S, h),$$

so the core of H is exactly K_n. Quotienting by the core gives the stated group. Compatibility with ψ gives the transition map. □

Theorem 2.9 (Arithmetic affine group)**.** *Let $\rho_n : G_K \to \mathrm{Aut}(A_n)$ be the Galois action on the α^n-torsion and put*

$$\Gamma_n = \langle H_n, \rho_n(G_K) \rangle \subseteq \mathrm{Aut}(A_n).$$

Since every element of H is defined over K, the Galois action centralizes its image on A_n. Thus $H_n \triangleleft \Gamma_n$. The arithmetic Galois group of the generic level-n quotient cover is

$$A_n \rtimes \Gamma_n.$$

Its geometric subgroup is $A_n \rtimes H_n$, and the corresponding constant quotient is Γ_n/H_n.

Proof. Write

$$Q_n = \sum_{j=0}^{n-1} \alpha^j Q, \qquad \psi^n(S) = \alpha^n S + Q_n.$$

The relation $\alpha h = \vartheta(h)\alpha$ gives $h\alpha^j = \alpha^j \vartheta^{-j}(h)$. Since Q is H-fixed, each $\alpha^j Q$ and hence Q_n is H-fixed. It also gives the equivariance identity

$$\psi^n(hS) = \vartheta^n(h)\psi^n(S).$$

Choose a generic point P of G above the generic point of G/H, and choose R_n with

$$\psi^n(R_n) = P, \qquad \text{equivalently} \qquad \alpha^n R_n = P - Q_n.$$

The roots of the quotient equation are $\pi(R_n + T)$ with $T \in A_n$.

Let an arithmetic Galois element restrict to $\sigma \in G_K$. Since $\sigma(P)$ and P have the same image in G/H, generic freeness gives a unique $h_\sigma \in H$ such that $\sigma(P) = h_\sigma P$. Put

$$k_\sigma = \vartheta^{-n}(h_\sigma).$$

Then

$$\psi^n(k_\sigma R_n) = \vartheta^n(k_\sigma)\psi^n(R_n) = h_\sigma P = \sigma(P) = \psi^n(\sigma(R_n)).$$

The difference

$$c_\sigma = \sigma(R_n) - k_\sigma R_n$$

therefore lies in A_n. On the branch labels the action is

$$T \longmapsto k_\sigma^{-1} c_\sigma + k_\sigma^{-1} \rho_n(\sigma) T.$$

Only the image of k_σ in H_n acts faithfully on A_n. The geometric group already contains all translations and this image of H_n; composing with those elements removes the translation and deck factors and leaves the pure Tate action $\rho_n(\sigma)$. Since ϑ^{-n} permutes H, the resulting linear group is still $\langle H_n, \rho_n(G_K)\rangle$; quotienting by the geometric subgroup gives the constant quotient. □

The specialization theorem admits a useful index form. The surrounding Kummer theory for points on commutative algebraic groups is developed in [34, 18, 21]. Put

$$T_\alpha(G) = \varprojlim_{\alpha} A_n.$$

A fixed $h \in H$ does not act componentwise on this inverse limit: the transition maps twist the deck action by ϑ. The compatible infinite-level action is

$$\iota(h)\big((T_n)_{n\geq 1}\big) = \big(\vartheta^{-n}(h)T_n\big)_{n\geq 1}. \tag{2.4}$$

Indeed,

$$\alpha\,\vartheta^{-(n+1)}(h) = \vartheta^{-n}(h)\alpha,$$

so (2.4) respects the inverse-system transitions. Since ϑ is an automorphism, its image at level n is the same subgroup $H_n \leq \mathrm{Aut}(A_n)$ as before.

Let $\rho_\alpha : G_K \to \mathrm{Aut}(T_\alpha(G))$ be the inverse-limit Galois representation and define

$$\Gamma_H = \langle \iota(H), \rho_\alpha(G_K)\rangle \subseteq \mathrm{Aut}(T_\alpha(G)).$$

Let G_P be the infinite specialized affine image and put $N_P = G_P \cap T_\alpha(G)$.

Corollary 2.10 (Exact specialization index)**.** *Under the hypotheses of theorem 2.12,*

$$[T_\alpha(G) \rtimes \Gamma_H : G_P] = [\Gamma_H : \rho_\alpha(G_K)]\,[T_\alpha(G) : N_P].$$

In particular, the quotient specialization is open in the generic affine group exactly when the ordinary Kummer image is open in the Tate module.

Proof. Let T be a profinite abelian group, let Λ act continuously on T, and let $G \leq T \rtimes \Lambda$ be closed with projection Γ. Put $N = G \cap T$ and $K = T \rtimes \Gamma$. Projection gives a bijection of coset spaces $(T \rtimes \Lambda)/K \simeq \Lambda/\Gamma$. There is a second bijection

$$T/N \longrightarrow K/G, \qquad t + N \longmapsto (t, 1)G.$$

Indeed, given $(t, \eta) \in K$, choose $(b, \eta) \in G$. Then $(t, \eta)(b, \eta)^{-1} = (t - b, 1)$, so every coset has a translation representative. Two such representatives give the same coset precisely when their difference belongs to N. Thus

$$[T \rtimes \Lambda : G] = [\Lambda : \Gamma]\,[T : N].$$

These are cardinalities of coset spaces; no normality of Γ in Λ or of G in K is required.

Apply this identity with $T = T_\alpha(G)$, $\Lambda = \Gamma_H$, and $\Gamma = \rho_\alpha(G_K)$. The latter has finite index in Γ_H, because the additional deck image is finite and

commutes with it. A closed subgroup of a profinite group is open exactly when its index is finite. This also proves the openness equivalence. □

Proposition 2.11 (Affine fixed-point proportion)**.** *Let A be a finite abelian group and let $\Gamma \leq \mathrm{Aut}(A)$. For the natural affine action of $A \rtimes \Gamma$ on A,*

$$\mathrm{FPP}(A \rtimes \Gamma \curvearrowright A) = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \frac{1}{|\ker(1-\gamma)|}.$$

The same formula holds for a linear coset after averaging over that coset.

Proof. For a fixed linear part γ, the affine map $x \mapsto a + \gamma x$ has a fixed point precisely when $a \in (1-\gamma)A$. The proportion of translations with this property is $|(1-\gamma)A|/|A| = |\ker(1-\gamma)|^{-1}$. Averaging over the linear parts proves the formula. □

Theorem 2.12 (No loss under quotient specialization)**.** *Assume that there is a point $C \in G(K)^H$ satisfying $(1-\alpha)C = Q$. Translation by C,*

$$t_C : G \longrightarrow G, \qquad R \longmapsto R + C,$$

is defined over K and is H-equivariant, so it descends to an automorphism $\bar{t}_C : G/H \to G/H$. If $\bar{\alpha} : G/H \to G/H$ is the map induced by α, then

$$f \circ \bar{t}_C = \bar{t}_C \circ \bar{\alpha}.$$

Let $P \in G(K)$ have trivial H-stabilizer and put $x = \bar{t}_C(\pi(P)) = \pi(P + C)$. Then, for every $n \geq 1$,

$$K(f^{-n}(x)) = K(A_n, \alpha^{-n}P).$$

Here each side denotes the field generated by the full geometric fiber. Thus quotient specialization is exactly the usual Kummer–Tate specialization after the stated K-rational linearization.

Proof. The relation $(1-\alpha)C = Q$ gives $\psi \circ t_C = t_C \circ \alpha$. Since C is fixed by H, the translation t_C is H-equivariant, and the same identity descends to $f \circ \bar{t}_C = \bar{t}_C \circ \bar{\alpha}$. Because $\bar{t}_C$ is defined over K, it does not change the field generated by a geometric fiber. It is therefore enough to compute the fiber of $\bar{\alpha}^n$ over $\pi(P)$.

Choose R_n with $\alpha^n R_n = P$. This fiber is labeled by $T \mapsto \pi(R_n + T)$ for $T \in A_n$. If two labels agree, the intervening $h \in H$ fixes P after applying α^n, hence is trivial. An automorphism of $K(A_n, R_n)$ fixing every quotient label acts affinely on A_n; the same stabilizer argument kills both its translation and linear parts. Hence the field generated by the quotient labels is $K(A_n, \alpha^{-n}P)$. □

Remark 2.13. If a linearizing point C exists only after a field extension K'/K, the same conclusion holds after replacing K by K'; no descent of the linearization to K is asserted.

Corollary 2.14 (Complete affine-coset fiber distribution)**.** *Let X be a normal geometrically integral variety of positive dimension over $\mathbf{F}_q$, and let $\pi : Y \to X$ be finite and generically étale. Let $U \subseteq X$ be the maximal open over which π is finite étale. Assume that the geometric generic fiber is labeled by a finite abelian group A and that its arithmetic monodromy, as a permutation group, is the natural affine action of $A \rtimes \Gamma$ on A, where $\Gamma \leq \mathrm{Aut}(A)$. Assume its geometric monodromy is $A \rtimes H$ with $H \triangleleft \Gamma$. In particular, $\deg \pi = |A|$. Suppose the constant-field quotient Γ/H is cyclic of order c, and choose $\gamma \in \Gamma$ mapping to arithmetic Frobenius.*

For $m \geq 1$ and $j \geq 0$, put

$$N_{m,j} = \#\{x \in U(\mathbf{F}_{q^m}) : \#Y_x(\mathbf{F}_{q^m}) = j\}.$$

For each residue class $a \pmod c$ and $j > 0$,

$$\lim_{\substack{m\to\infty \\ m\equiv a \pmod c}} \frac{N_{m,j}}{\#U(\mathbf{F}_{q^m})} = \frac{1}{|H|} \sum_{\substack{h\in H \\ |\ker(1-\gamma^a h)|=j}} \frac{1}{j}. \tag{2.5}$$

The limiting mass at $j = 0$ is the complementary mass. Including the base points in $X \setminus U$ does not change these limits.

Proof. The rational points of an étale fiber are the fixed points of its arithmetic Frobenius permutation. For $m \equiv a \pmod c$, these permutations lie in the coset

$$G_a = \{(b, \gamma^a h) : b \in A,\ h \in H\}, \qquad |G_a| = |A||H|.$$

Let $C_{a,j} \subseteq G_a$ consist of the permutations with exactly j fixed points. This subset is conjugacy invariant. For $L = \gamma^a h$, write $k_L = |\ker(1 - L)|$. The fixed-point equation is

$$(1 - L)z = b.$$

Every nonempty solution set is a coset of $\ker(1 - L)$, so it has k_L elements. Exactly $|A|/k_L$ translations b make the equation soluble. Consequently, for $j > 0$,

$$|C_{a,j}| = \sum_{\substack{h\in H \\ k_{\gamma^a h}=j}} \frac{|A|}{j}.$$

Finite-field Chebotarev, with the constant-field Frobenius coset fixed, gives

$$N_{m,j} = \frac{|C_{a,j}|}{|A||H|}\, q^{m \dim X} + O(q^{m(\dim X - 1/2)}).$$

See [9, Theorem 3] and [11]. The constant depends on the fixed cover, not on m. One may restrict to a dense affine open; the omitted locus has smaller dimension. After extension of constants to $\mathbf{F}_{q^m}$, the arithmetic image is the preimage of $\langle \gamma^m H \rangle$; its relevant Frobenius coset is still G_a, so the denominator is the geometric-group order $|A||H|$. Lang–Weil gives the corresponding leading term for $\#U(\mathbf{F}_{q^m})$ [20]. Dividing and using the preceding count proves (2.5).

There are only finitely many possible fixed-point counts, proving the complementary-mass assertion. Finally $X\backslash U$ has dimension at most $\dim X-1$ and contributes only $O(q^{m(\dim X-1)})$ base points. □

Remark 2.15 (The monodromy action is part of the hypothesis). Abstract isomorphisms of the geometric and arithmetic groups with $A \rtimes H$ and $A \rtimes \Gamma$ do not suffice. The fixed-point calculation uses their specified permutation action on the sheets, identified with A.

2.4. **Orbit Witt vectors and quotient transfer.** The affine fixed-point formula records only the first question about a permutation, namely whether a point is fixed. For quotient towers it is useful to retain every cycle length. Let S be a finite set and let φ be a permutation of S. The corresponding rational Witt construction is classical [1, 6, 16]. Define

$$Z_{S,\varphi}(t) = \det(1 - t\varphi \mid \mathbb{Z}[S])^{-1} \in 1 + t\mathbb{Z}[[t]]. \tag{2.6}$$

For $f(t) \in 1 + tR[[t]]$, write

$$w_r(f) = [t^r]\, t\frac{f'(t)}{f(t)}.$$

Proposition 2.16 (Orbit Witt classification)**.** *If $c_d(\varphi)$ is the number of d-cycles of φ, then*

$$Z_{S,\varphi}(t) = \prod_{d\geq 1}(1 - t^d)^{-c_d(\varphi)}, \tag{2.7}$$

$$w_r(Z_{S,\varphi}) = |\operatorname{Fix}(\varphi^r)|. \tag{2.8}$$

The cycle counts are recovered by Möbius inversion:

$$c_d(\varphi) = \frac{1}{d}\sum_{e\mid d}\mu(d/e)|\operatorname{Fix}(\varphi^e)|. \tag{2.9}$$

Proof. A cycle of length d contributes the factor $1 - t^d$ to the characteristic polynomial of the permutation lattice. Multiplying over the cycles proves (2.7). Logarithmic differentiation gives

$$t\frac{d}{dt}\log Z_{S,\varphi}(t) = \sum_{r\geq 1}\left(\sum_{d\mid r} d c_d(\varphi)\right) t^r.$$

The coefficient in parentheses counts the points fixed by φ^r. Formula (2.9) is the inverse divisor relation. □

The assignment in (2.6) is the finite rational part of the Burnside–Witt correspondence of Dress and Siebeneicher [6]; compare the necklace description of Metropolis and Rota [26]. Disjoint union becomes multiplication of power series, which is addition in the big Witt ring, and Cartesian product becomes Witt multiplication because fixed-point counts multiply.

For an affine permutation $\varphi_{a,\gamma}(x) = a + \gamma x$ of a finite abelian group A, put $N_r(\gamma) = 1 + \gamma + \cdots + \gamma^{r-1}$. Iteration gives

$$\varphi^r_{a,\gamma}(x) = N_r(\gamma)a + \gamma^r x.$$

The fixed-point equation is therefore a linear equation in A.

Proposition 2.17 (Affine orbit ghosts)**.** *For every $r \geq 1$,*

$$w_r(Z_{A,\varphi_{a,\gamma}}) = \begin{cases} |\ker(1-\gamma^r)|, & N_r(\gamma)a \in (1-\gamma^r)A, \\ 0, & N_r(\gamma)a \notin (1-\gamma^r)A. \end{cases}$$

Together with (2.9)*, this determines the complete cycle inventory of the affine permutation.*

The orbit Witt vector also has an exact composition law. Suppose $\pi : S \to T$ is a surjection of finite sets and φ, ψ are permutations with $\pi\varphi = \psi\pi$. For a cycle C of ψ, choose $t_C \in C$, put $d_C = |C|$, and let

$$\rho_C = \varphi^{d_C}|_{\pi^{-1}(t_C)}$$

be the return permutation. Changing t_C conjugates ρ_C.

Theorem 2.18 (Orbit-Witt transfer)**.** *With the preceding notation,*

$$Z_{S,\varphi}(t) = \prod_{C \in \mathrm{Cyc}(\psi)} Z_{\pi^{-1}(t_C),\rho_C}(t^{d_C}). \tag{2.10}$$

Equivalently,

$$|\operatorname{Fix}(\varphi^r)| = \sum_{\substack{C \in \mathrm{Cyc}(\psi) \\ d_C \mid r}} d_C \, |\operatorname{Fix}(\rho_C^{r/d_C})|.$$

Proof. A cycle of length e for the return map over C produces one φ-cycle of length $d_C e$. Indeed, the point must travel once around the base cycle before the return map is applied, and it returns after exactly e circuits, giving (2.10). Taking ghost coordinates gives the fixed-point formula. □

If $Y/\mathbb{F}_q$ is finite étale and Frobenius acts on $S = Y(\overline{\mathbb{F}}_q)$, then Z_{S,Frob_q} is the Hasse–Weil zeta function of Y; see [27, Chapter V]. When Y is the root scheme of a squarefree polynomial, Frobenius cycles are its irreducible factors. The orbit-Witt transfer thus gives a recursive formula for every factor degree in a finite quotient tower. The fixed-point distribution in corollary 2.14 is the first layer of this more refined cycle invariant.

2.5. **Discriminant divisors as a semigroup cocycle.** The determinant line also records ramification before passing to a square class; the relation between the different and discriminant is reviewed in [38, Tag 0BW8] and [35, Chapter III]. Let X be a smooth integral Noetherian scheme and let $f : X \to X$ be finite flat, generically étale, and Gorenstein. Write R_f for the different divisor on the source and $\Delta_f = f_* R_f$ for the discriminant divisor on the target, transported to the same copy of X.

Theorem 2.19 (Integral discriminant cocycle)**.** *For two such maps f and g,*

$$R_{f\circ g} = R_g + g^* R_f, \tag{2.11}$$

$$\Delta_{f\circ g} = (\deg g)\Delta_f + f_*\Delta_g. \tag{2.12}$$

The operators

$$\mathcal{B}_f = \begin{pmatrix} f_* & \Delta_f \\ 0 & \deg f \end{pmatrix}$$

form a semigroup representation on $\mathrm{Div}(X) \oplus \mathbb{Z}$.

Proof. For a tower of finite Gorenstein maps, the relative dualizing modules satisfy

$$\omega_{Y/X} \simeq \omega_{Y/Z} \otimes p^* \omega_{Z/X}.$$

After identifying the inverse dualizing modules with the different ideals, this becomes (2.11). Push forward along $f \circ g$. The first term gives $f_*\Delta_g$, while the projection formula turns the second into $(\deg g)\Delta_f$, yielding (2.12). The matrix identity follows by direct multiplication using $(f \circ g)_* = f_* g_*$. □

Suppose a finite set of prime divisors $\mathcal{P}$ contains the support of each discriminant divisor and is forward invariant under the maps under consideration. Then the operators $\mathcal{B}_f$ preserve the lattice $\mathrm{Div}_{\mathcal{P}}(X) \oplus \mathbb{Z}$. Reducing the matrices modulo an integer gives a finite automaton for the ramification multiplicities of every composition word. For one map of degree m, iteration of (2.12) gives

$$\Delta_{f^r} = \sum_{j=0}^{r-1} m^{r-1-j} f_*^j \Delta_f.$$

Reduction of (2.12) modulo two records only the parity of the coefficients of the discriminant divisor Δ_f. When 2 is invertible, this does not determine the sign character of the étale monodromy permutation representation: passage from the determinant section of the trace pairing in $\Gamma(X, L_f^{-2})$ to its divisor forgets nonsquare unit factors. Thus the divisor cocycle retains all discriminant multiplicities, whereas the determinant section is required to recover the sign character.

3. Norm-coherent quotient equations and higher jets

Let F be the universal deformation of a one-dimensional formal group of height h over the Lubin–Tate deformation ring [24, 14]

$$\mathcal{O}_h = W(k)[[u_1, \ldots, u_{h-1}]].$$

3.1. **Norm-coherent coordinates.** In this subsection assume that k is an algebraic extension of $\mathbf{F}_p$. Fix a finite-height formal group Γ/k, and a coordinate x_0 on Γ. A *coordinate deformation* over a complete local ring $(R, \mathfrak{m}_R)$ consists of a deformation F/R of Γ, a chosen identification of its special fiber with Γ, and a coordinate X reducing to x_0. Coordinate changes are Zhu's $\star$-isomorphisms: isomorphisms compatible with the fixed special-fiber identifications. Their linear coefficients need not be exactly 1.

Theorem 3.1 (Zhu's norm-coherent representative)**.** *Under the preceding algebraicity hypothesis, every $\star$-isomorphism class of coordinate deformations contains a unique norm-coherent representative. For that representative and every finite subgroup $D \subset F$, the quotient coordinate X_D is characterized by*

$$X_D = N_D(X), \qquad q_D^* X_D = q_D^* N_D(X) = \prod_{a \in D} (X +_F a). \tag{3.1}$$

The construction is preserved by complete local base change and by pushforward along isogenies; in particular, the coordinates selected along a chain of subgroup quotients are compatible with norm transitivity.

Proof. Existence and uniqueness in the coordinate formulation are [46, Proposition 7.1 and Theorem 7.22]. The preservation of norm coherence under complete local base change and pushforward along an isogeny is [46, Proposition 6.22]. For a chain $D \subset D'$, finite-free norm transitivity gives

$$N_{D'}(X) = N_{D'/D}\big(N_D(X)\big),$$

so the target coordinate obtained in one step agrees with the coordinate obtained through F/D. This is precisely the chain compatibility used below. □

Norm coherence makes the higher prepared coefficients compatible along the quotient tower. They depend on the initial coordinated deformation; the algebraicity hypothesis is used only for this coherent choice, not for the determinant norm or the trace–Euler argument.

3.2. **Resultant reconstruction.** We use the standard determinant and root descriptions of resultants; see [13].

Theorem 3.2 (All-order resultant reconstruction)**.** *Let*

$$P_\epsilon(T, Y) = P_0(T, Y) + \sum_{a=1}^{N} \epsilon^a A_a(T, Y)$$

be monic of degree q in T, and write

$$\mathrm{Res}_T(P_\epsilon, g) = \sum_{n=0}^{N} \epsilon^n R_n(g) \pmod{\epsilon^{N+1}}.$$

Put $A_0 = P_0$. If g has unit leading coefficient, pass to a faithfully flat splitting of $b^{-1}g$ and write $g(T) = b\prod_{j=1}^{m}(T - \beta_j)$. Then

$$R_n(g) = (-1)^{qm} b^q \sum_{\nu_1 + \cdots + \nu_m = n} \prod_{j=1}^{m} A_{\nu_j}(\beta_j, Y).$$

In particular,

$$R_n(T - Z) = (-1)^q A_n(Z, Y).$$

Hence the linear-probe norm jets recover the complete prepared quotient equation, and that equation recovers every polynomial norm.

Proof. First suppose that the leading coefficient of g is a unit. A monic polynomial admits a finite faithfully flat splitting algebra, and the root formula there gives

$$\mathrm{Res}_T(P_\epsilon, g) = (-1)^{qm} b^q \prod_{j=1}^{m} \left(\sum_{a=0}^{N} \epsilon^a A_a(\beta_j, Y) \right) \pmod{\epsilon^{N+1}}.$$

The coefficient of ϵ^n is obtained by choosing indices $\nu_1, \dots, \nu_m$ whose sum is n, which is the stated formula. The expression is symmetric in the roots of g. It therefore belongs to the original algebra and descends from the splitting extension.

For a probe with arbitrary leading coefficient, take the universal coefficient ring over $\mathbf{Z}$ for P_ϵ and g. The resultant coefficients are polynomials in these coefficients, computed by the Sylvester determinant. The identity just proved holds after inverting the universal leading coefficient. This localization is injective, so the resulting polynomial identity holds in the universal ring and can be specialized to every coefficient ring. No faithfully flat extension making a nonunit leading coefficient invertible is required.

The formula is triangular with respect to the prepared corrections. Terms containing A_n use it at exactly one root and use A_0 at all remaining roots; every other term involves only $A_1, \dots, A_{n-1}$. For the linear probe $g = T - Z$ there is only one root, so no lower-order cross term occurs and

$$R_n(T - Z) = (-1)^q A_n(Z, Y).$$

Taking Z to be an indeterminate recovers the polynomial A_n coefficient by coefficient. An induction on n recovers the full prepared equation. Once that equation is known, the norm of any polynomial probe is its resultant with the equation, which proves the final assertion. □

3.3. **Composition and directional norm orders.** Norm transitivity gives coefficient substitution identities of Hasse–Schmidt type; compare [15]. A norm is a multiplicative polynomial law, not in general an additive higher derivation; the distinction matters for return orders. We retain the exact chain law and isolate its primary contribution.

Theorem 3.3 (Chain law and primary special-fiber contribution)**.** *For composable quotient norms $N_{i,\varepsilon} = \sum_a \varepsilon^a H_{i,a}$,*

$$N_{21,\varepsilon}(x) = N_{2,\varepsilon}(N_{1,\varepsilon}(x)).$$

Suppose that in characteristic p the outer primary component $H_{2,0}$ is the qth-power map, where q is a power of p. On an inner input $u + \varepsilon^b v + O(\varepsilon^{b+1})$, its primary contribution is

$$H_{2,0}(u) + \varepsilon^{qb} v^q + O(\varepsilon^{q(b+1)}).$$

This assertion concerns only $H_{2,0}$. Terms $\varepsilon^a \big(H_{2,a}(u + \varepsilon^b v + \cdots) - H_{2,a}(u)\big)$, $a > 0$, can contribute below order qb, even in characteristic p.

Proof. For finite locally free algebras $A \to B \to C$, determinant norms satisfy $N_{C/A} = N_{B/A} \circ N_{C/B}$. Apply this identity over each truncated coefficient ring and extract coefficients. The primary assertion is the Frobenius identity $(u + \varepsilon^b v + \cdots)^q = u^q + \varepsilon^{qb} v^q + \cdots$. It places no restriction on the higher outer components. □

Remark 3.4 (A finite-free counterexample to full return). Over $A = \mathbf{F}_3[[\varepsilon]]$, let $B = A[t]/(t^3 - \varepsilon t)$. It is finite free of rank three, generically étale, and its special-fiber norm is the cube map. Nevertheless,

$$N_{B/A}(t + a) = a^3 - \varepsilon a, \qquad N_{B/A}(t + \varepsilon) - N_{B/A}(t) = \varepsilon^3 - \varepsilon^2.$$

The full perturbation first appears at order two, not three.

Proposition 3.5 (Directional determinant bound)**.** *Let $A = k[[\varepsilon]]$ and let B be a finite free A-algebra of rank q. For $u, v \in B$, define $c_i \in A$ by*

$$\det(m_u + Z m_v) = \sum_{i=0}^{q} c_i Z^i.$$

For every $b \geq 1$,

$$N(u+\varepsilon^b v) - N(u) = \sum_{i=1}^{q} \varepsilon^{bi} c_i, \qquad v_\varepsilon\big(N(u+\varepsilon^b v) - N(u)\big) \geq \min_{1 \leq i \leq q} \{bi + v_\varepsilon(c_i)\}. \tag{3.2}$$

Here $v_\varepsilon(0) = \infty$; equality holds when the finite minimum is unique. If the reduced norm is a qth-power law in characteristic p, with q a power of p, then $c_i \in \varepsilon A$ for $0 < i < q$, and the order is at least $\min\{qb, b + 1\}$. In particular, if c_q is a unit and $bi + v_\varepsilon(c_i) > qb$ for all $i < q$, the order is exactly qb.

Proof. Multiplication is A-linear, so $m_{u+Zv} = m_u + Z m_v$. Substitute $Z = \varepsilon^b$ in the determinant polynomial and subtract its constant term. The valuation inequality follows from the ultrametric inequality; a unique least-order summand cannot cancel. Under the reduced power-law hypothesis all intermediate coefficients vanish modulo ε, while $c_q = N(v)$. These facts give the remaining bounds. □

For example, in characteristic p the rank-q algebra $A[t]/(t^q - \varepsilon^a t)$ has $N(t + \varepsilon^b) = \varepsilon^{qb} - \varepsilon^{a+b}$. When the two exponents are distinct the minimum is attained, showing that mixed terms can give the sharp earlier order. Equal exponents can cancel, as the proposition allows.

3.4. **Subgroup width and ordered branches.** The subgroup numbers are Hall polynomials, equivalently Gaussian binomial coefficients; see [3, 25].

Definition 3.6 (Scalar Witt streams)**.** *Let $\mathscr{C}_E$ denote the graded polynomial algebra corepresenting the Lubin–Tate–Witt functor. A scalar Witt stream is a sequence of homogeneous elements $z_d \in (\mathscr{C}_E)_{p^d}$, one at each depth. A family of M streams is* indecomposably faithful *if, for every d, the classes of its M depth-d elements span the degree-p^d indecomposable quotient of $\mathscr{C}_E$.*

This is the precise bounded-width notion used below; no assertion is made about arbitrary forms of encoding.

Theorem 3.7 (Subgroup width)**.** *The number of order-p^d subgroups of $(\mathbf{Q}_p/\mathbf{Z}_p)^h$ is*

$$N_{h,d} = \begin{bmatrix} d+h-1 \\ h-1 \end{bmatrix}_p, \qquad \sum_{d\geq 0} N_{h,d} T^d = \prod_{j=0}^{h-1} (1-p^j T)^{-1}.$$

The depth-d indecomposable layer of $\mathscr{C}_E$ has rank $N_{h,d}$. If $h > 1$, no fixed finite family of scalar Witt streams is indecomposably faithful.

Proof. The graph-of-a-homomorphism count in section A proves

$$N_{h,d} = \sum_{a=0}^{d} p^{a(h-1)} N_{h-1,d-a}, \qquad N_{1,d} = 1.$$

Multiplying generating series gives the displayed product, whose coefficients are the Gaussian binomial coefficients. Strickland's subgroup count and the polynomial-corepresenter calculation identify this number with the rank of $Q(\mathscr{C}_E)_{p^d}$ [39, 33].

A family of M streams supplies at most M classes in that degree. Spanning a free module of rank $N_{h,d}$ forces $M \geq N_{h,d}$, as is seen after reduction modulo the maximal ideal. For $h > 1$ the generating function gives $N_{h,d} \geq p^{(h-1)d}$, so these ranks are unbounded. □

Remark 3.8 (Digit order in a chosen Tate trivialization)*.* Let $T \simeq \mathbf{Z}_p^h$, put $V = T/pT$, and choose representatives $s : V \to T$ with $s(0) = 0$. Successive reduction and division by p give a bijection of sets

$$V^r \longrightarrow T/p^r T, \qquad (d_0, \ldots, d_{r-1}) \longmapsto \sum_{i=0}^{r-1} p^i s(d_i) \pmod{p^r T}.$$

To prove existence and uniqueness, reduce a class modulo p, subtract its chosen representative, divide the difference by p, and repeat. When inverse branches have been labeled by these successive choices, chronological order records the lowest-stage digit first. Reading the same expansion in the order of the subquotients

$$p^{r-1}T/p^r T,\ p^{r-2}T/p^{r-1}T, \ldots, T/pT$$

reverses the list of digits. This is a convention for the chosen coordinates, not a canonical additive map on $T/p^r T$. In rank one at depth two with the usual representatives, reversal sends 1 to p; their additive orders differ, so reversal is not even a group automorphism in this case.

At height one, the width is one at every depth and a single p-typical stream is indecomposably faithful. At height two, $N_{2,d} = 1 + p + \cdots + p^d$; this growth is already visible in the endpoint divisors of section 9.

3.5. **Characteristic norms and Hankel matrices.** The conventions for characteristic polynomials, symmetric functions, and subdiscriminants follow [25, 13]. The determinant norm contains more than its top coefficient. Let B be a finite projective A-algebra of constant rank n and let $x \in B$. Multiplication by x is an endomorphism m_x of a finite projective module, and

$$\begin{aligned} N_{B/A}(t-x) &= \det(t - m_x) \\ &= \sum_{j=0}^{n} (-1)^j \operatorname{Tr}(\wedge^j m_x) t^{n-j}. \end{aligned} \tag{3.3}$$

The two extreme nonleading coefficients are

$$\begin{aligned} [t^{n-1}]N_{B/A}(t-x) &= -\operatorname{Tr}_{B/A}(x), \\ [t^0]N_{B/A}(t-x) &= (-1)^n N_{B/A}(x). \end{aligned}$$

If $B = A[x]$ has monic defining polynomial P, then $P(t) = N_{B/A}(t-x)$.

Package the power traces in the big Witt ring by setting

$$\Xi_{B/A}(x;t) = N_{B/A}(1-tx)^{-1} \in 1 + tA[[t]]. \tag{3.4}$$

Theorem 3.9 (Characteristic-Witt ghost formula)**.** *For $r \geq 1$,*

$$w_r(\Xi_{B/A}(x;t)) = \operatorname{Tr}_{B/A}(x^r). \tag{3.5}$$

More generally, finite-flat norm on power series defines a map $N^W_{B/A} : W(B) \to W(A)$ whose ghost coordinates are obtained by applying $\operatorname{Tr}_{B/A}$, and these maps are transitive in towers.

Proof. Let M_f be multiplication by f on the finite projective $A[[t]]$-module $B[[t]]$. Jacobi's formula gives

$$\frac{d}{dt} \log \det(M_f) = \operatorname{Tr}_{B/A}(f'/f).$$

Apply this to $f = (1-tx)^{-1}$ and compare coefficients. The general statement follows in the same way, while transitivity is the ordinary transitivity of finite locally free norms after extension to power-series rings. □

Assume now that $B = A[x]$ is free with basis $1, x, \ldots, x^{n-1}$, and write $s_r = \operatorname{Tr}_{B/A}(x^r)$, $s_0 = n$. Let

$$H_x = (s_{i+j})_{0 \leq i,j < n}$$

be the Hankel moment matrix.

Theorem 3.10 (Witt–Hankel reconstruction)**.** *Write*

$$N_{B/A}(1-tx) = 1 + d_1 t + \cdots + d_n t^n.$$

Then the moments satisfy

$$s_{r+n} + d_1 s_{r+n-1} + \cdots + d_n s_r = 0 \qquad (r \geq 0),$$

and H_x is the Gram matrix of the trace pairing in the power basis. In particular,

$$\operatorname{disc}(B/A) = \det H_x. \tag{3.6}$$

For $1 \le m \le n$, the leading $m \times m$ minor is the mth subdiscriminant. After a splitting extension with sheets $x_1, \dots, x_n$, it is

$$\sum_{|S|=m} \prod_{i<j \in S} (x_i - x_j)^2.$$

Proof. The polynomial $z^n + d_1 z^{n-1} + \cdots + d_n$ is the characteristic polynomial of m_x. Cayley–Hamilton, multiplied by m_x^r and followed by trace, gives the recurrence. The (i, j) entry of the trace pairing in the power basis is $\operatorname{Tr}(x^{i+j}) = s_{i+j}$, proving (3.6). Over a splitting algebra, the leading Hankel block is $V^t V$ for the corresponding Vandermonde matrix. The subdiscriminant formula follows from Cauchy–Binet. □

Let A be a DVR and suppose the trace pairing is generically nondegenerate. The determinantal ideals of H_x are the Fitting ideals of the trace-dual defect $\operatorname{coker}(B \to B^\vee)$. Hence the characteristic Witt vector determines the elementary divisors of the codifferent. In a monogenic complete intersection $B = A[T]/(W)$, the different is generated by $\partial_T W$. The higher resultant jets from theorem 3.2 recover every coefficient of W and therefore recover the complete different and discriminant profile.

Proposition 3.11 (Trace-dual Fitting profile)**.** *Let A be a DVR and let B be finite free over A, with generically nondegenerate trace pairing. If*

$$\lambda : B \longrightarrow B^\vee, \qquad b \longmapsto (b' \mapsto \operatorname{Tr}_{B/A}(bb')),$$

has Smith exponents $a_1 \le \cdots \le a_n$, then the ideal generated by its $r \times r$ minors has valuation $a_1 + \cdots + a_r$. In a power basis these ideals are generated by the corresponding minors of the Hankel matrix in theorem 3.10. The top ideal is the discriminant ideal.

Proof. Choose Smith bases for λ. The matrix is diagonal with entries $\pi^{a_1}, \dots, \pi^{a_n}$ up to multiplication by invertible matrices. The least valuation among the $r \times r$ minors is the sum of the r smallest exponents, and invertible row and column operations do not change the determinantal ideal. The Hankel description follows from the power basis computation above. □

4. Fixed-source endpoint algebras

The universal subgroup scheme has source and target maps to the deformation space [22, 39]. We fix the source over $W(k)$ and retain the target, obtaining the finite endpoint algebra used below.

Fix a perfect field k of characteristic p, a one-dimensional formal group Γ/k of finite height h, and its universal deformation ring

$$\mathcal{O} = W(k)[[u_1, \dots, u_{h-1}]].$$

Let $\mathcal{O}_d$ denote the Strickland ring classifying subgroup schemes of order p^d in the universal deformation [41, Theorem 1.1]. It carries source and target maps

$$s_d, t_d : \mathcal{O} \longrightarrow \mathcal{O}_d.$$

Choose a mixed-characteristic lift of Γ and parameters for which its classifying section is $\epsilon_W(u_i) = 0$. There are two different maps:

$$\epsilon_W : \mathcal{O} \longrightarrow W(k), \qquad \epsilon_k : \mathcal{O} \xrightarrow{\epsilon_W} W(k) \longrightarrow k.$$

The first fixes the source lift; the second is the residue map. The parameter-zero section is a choice, not a canonical lift of an arbitrary Γ. We use "Honda source" only when a specified Honda model and its chosen lift have been identified. Define

$$B_d = \mathcal{O}_d \widehat{\otimes}_{\mathcal{O}, s_d, \epsilon_W} W(k), \qquad \tau_d : \mathcal{O} \longrightarrow B_d, \qquad E_d = \ker(\tau_d).$$

Remark 4.1 (Source coefficients and Frobenius). Let φ denote Witt Frobenius on $W(k)$. We always regard B_d as a $W(k)$-algebra through the source section. The target coefficient action is

$$\tau_d(a) = \varphi^d(a) \qquad (a \in W(k)),$$

where the right side is in the source coefficient copy. All finite-free traces, duals, and lengths below use that source structure. In particular, we do not untwist the traces independently at different depths when forming one Witt ghost sequence. A fixed-degree Frobenius twist preserves module lengths and elementary divisors, but not the compatibility of an arbitrary sequence with Witt ghost congruences. Recognition uses ϵ_k; the integral trace argument uses ϵ_W.

Put

$$N_{h,d} = \#\{D \le (\mathbf{Q}_p/\mathbf{Z}_p)^h : |D| = p^d\} = \begin{bmatrix} d+h-1 \\ h-1 \end{bmatrix}_p, \qquad L_d = \sum_{j=0}^{d-1} N_{h,j}.$$

Theorem 4.2 (All-height geometric Witt recursion)**.** *For every finite height h and every $d \ge 0$,*

$$I_{d+1} = pI_d + E_d.$$

Equivalently, if $J_0 = \mathcal{O}$ and $J_{d+1} = pJ_d + E_d$, then $J_d = I_d$ for all d.

Proposition 4.3 (Integral endpoint algebra)**.** *The target map τ_d is surjective, and*

$$\mathcal{O}/E_d \simeq B_d.$$

The ring B_d is finite free and Gorenstein over $W(k)$ of rank $N_{h,d}$. In particular,

$$E_d \cap p\mathcal{O} = pE_d.$$

Moreover,

$$\frac{E_d + p\mathcal{O}}{p\mathcal{O}} = \frac{I_{d+1} + p\mathcal{O}}{p\mathcal{O}}.$$

Proof. Strickland's theorem makes $\mathcal{O}_d$ finite free and Gorenstein over the source copy of $\mathcal{O}$ [41, Theorem 1.1]. The sequence $u_1, \dots, u_{h-1}$ is a regular sequence in $\mathcal{O}_d$, and source specialization gives a finite free Gorenstein $W(k)$-algebra of the stated rank. Rezk's characteristic-p endpoint theorem [33, Propositions 8.1–8.5] identifies the reduction of τ_d with the surjective target map

$$\mathcal{O}/p \longrightarrow \mathcal{O}_d \otimes_{\mathcal{O},s_d} k$$

and identifies its kernel with the reduction of I_{d+1}. The cokernel of τ_d is a finite $W(k)$-module with zero reduction, so Nakayama's lemma makes τ_d surjective. Finally, B_d is p-torsion-free, which is equivalent to the saturation identity for E_d. □

Corollary 4.4 (The height-three endpoint ideals)**.** *When $h = 3$, every E_d is a two-generator complete-intersection ideal. There are elements $F_d, G_d \in \mathcal{O}$ such that*

$$E_d = (F_d, G_d), \qquad I_{d+1} = pI_d + (F_d, G_d).$$

Proof. The regular local ring $W(k)[[u_1, u_2]]$ has dimension three, while B_d is finite flat over the one-dimensional DVR $W(k)$. Hence E_d has grade two, is perfect, and $\mathcal{O}/E_d$ has Cohen–Macaulay type one. If $\mu(E_d)$ denotes its minimal number of generators, the Hilbert–Burch resolution [8, Chapter 20] of a grade-two perfect ideal has last free module of rank $\mu(E_d) - 1$; this rank equals the type of the quotient. Thus $\mu(E_d) - 1 = 1$, so E_d has two generators and is a complete intersection. □

4.1. **The geometry in small height.** The codimension of E_d is $h - 1$, so the form of the endpoint ideal changes with the height. At height one there are no deformation parameters: $\mathcal{O} = W(k)$, the fixed-source target algebra is $W(k)$ up to the harmless coefficient Frobenius, and $E_d = 0$. The recursion of theorem 4.2 reduces to the classical filtration $I_d = p^d W(k)$.

At height two, E_d has grade one in the regular local ring $W(k)[[u_1]]$. The quotient is finite flat over $W(k)$, so E_d is generated by a monic Weierstrass polynomial H_d. The all-height theorem gives

$$I_{d+1} = pI_d + (H_d).$$

The separate Hecke argument in section 9 identifies the interaction of consecutive generators and proves the stronger formula $I_d = (H_{d-1}, H_d)$.

At height three the Gorenstein property already forces a complete intersection. In higher codimension this instead follows from the Euler presentations and the minimal-cone argument of theorem 8.3.

4.2. **The endpoint Bockstein and its trace bound.** The endpoint ideals are nested after reduction modulo p. For $d \geq 1$,

$$E_d \subseteq E_{d-1} + p\mathcal{O}.$$

Since B_{d-1} is p-torsion-free, division by p modulo E_{d-1} is well-defined.

Definition 4.5 (Endpoint Bockstein)**.** *For $d \geq 1$, define*

$$\beta_d : E_d \longrightarrow B_{d-1}, \qquad \tau_{d-1}(x) = p\beta_d(x).$$

For a finite free $W(k)$-algebra B, put

$$\mathcal{T}^a(B) = \{b \in B : \mathrm{Tr}_{B/W(k)}(bB) \subseteq p^a W(k)\}.$$

Lemma 4.6 (Trace-ghost normalization and variance)**.** *Let $P_{\mathrm{tot}} : \mathcal{O} \to \mathbb{W}_E(\mathcal{O})$ be the total E-typical power operation. Define*

$$\xi = \mathrm{pr}_p \circ \mathrm{tr} \circ \mathrm{sp}_{\epsilon_W} \circ P_{\mathrm{tot}} : \mathcal{O} \longrightarrow W_p(W(k)), \tag{4.1}$$

where sp_{ϵ_W} is base change along the fixed-source map, tr *is Rezk's natural transformation from E-typical to big Witt vectors, and pr_p is projection to the p-typical factor. These are covariant natural transformations in the coefficient algebra.*

In weight p^n, source specialization uses s_n, while the total operation uses t_n. With the source convention of remark 4.1, the ordinary, unnormalized trace satisfies

$$w_n(\xi(z)) = \mathrm{Tr}^{\mathrm{source}}_{B_n/W(k)}(\tau_n(z)) \qquad (z \in \mathcal{O}). \tag{4.2}$$

For a coefficient $a \in W(k)$ this reads $w_n(\xi(a)) = N_{h,n}\varphi^n(a)$. In particular, the first ghost congruence is $w_1 \equiv w_0^p \pmod p$, as required. No degreewise untwisting is used. The formula is preserved by further complete local base change.

Proof. The p^n-component of $\mathrm{sp}_{\epsilon_W} P_{\mathrm{tot}}(z)$ is the image of $t_n(z)$ in

$$B_n = \mathcal{O}_n \widehat{\otimes}_{\mathcal{O}, s_n, \epsilon_W} W(k),$$

namely $\tau_n(z)$. Rezk's trace-ghost formula identifies the p^n-ghost of the composite (4.1) with the ordinary finite-free algebra trace of this element [33, Propositions 11.1 and 11.7–11.11]. Naturality of the total operation, source base change, the Witt trace, and the p-typical projection gives the final assertion. □

Theorem 4.7 (Trace–Bockstein bound)**.** *Let $x \in E_s$. If $n \neq s$, then*

$$\tau_n(x) \in \mathcal{T}^{\min(n,s)+1}(B_n),$$

while $\tau_s(x) = 0$. In particular,

$$\beta_d(E_d) \subseteq \mathcal{T}^{d-1}(B_{d-1}).$$

Proof. Fix $y \in \mathcal{O}$, put $z = xy$, and write

$$\xi(z) = (a_0, a_1, \ldots) \in W_p(W(k)).$$

By lemma 4.6,

$$\mathrm{Tr}_{B_n/W(k)}(\tau_n(z)) = w_n(\xi(z)) = \sum_{i=0}^{n} p^i a_i^{p^{n-i}}. \tag{4.3}$$

Since $x \in E_s$, one has $\tau_s(x) = 0$, and hence $w_s(\xi(z)) = 0$.

The zeroth Witt coordinate is the source augmentation, so $a_0 = \epsilon_W(z)$. The equality

$$\frac{E_s + p\mathcal{O}}{p\mathcal{O}} = \frac{I_{s+1} + p\mathcal{O}}{p\mathcal{O}}$$

from proposition 4.3, together with $I_{s+1} \subseteq I_1 = (p, u_1, \dots, u_{h-1})$, gives $\epsilon_W(x) \in pW(k)$. Thus $a_0 = \epsilon_W(x)\epsilon_W(y) \in pW(k)$. For $s = 0$ one has $E_0 = (u_1, \dots, u_{h-1})$, so $a_0 = 0$.

Suppose that some a_j, $j \leq s$, is not divisible by p, and choose the least such j. In w_s, the jth summand has valuation exactly j. Every term with index greater than j has valuation at least $j+1$, while for $i < j$ minimality gives

$$v_p(p^i a_i^{p^{s-i}}) \geq i + p^{s-i} \geq j + 1.$$

This contradicts $w_s = 0$. Thus $a_0, \dots, a_s \in pW(k)$. Substitution in (4.3) gives

$$w_n(\xi(z)) \in p^{\min(n,s)+1} W(k) \qquad (n \neq s).$$

As y varies, the surjection τ_n makes $\tau_n(y)$ range over B_n. The displayed divisibility therefore holds after multiplying $\tau_n(x)$ by every element of B_n, which is exactly the trace-lattice inclusion. Taking $n = d-1$ and cancelling one factor of p gives the Bockstein statement. □

5. Primitive duality and trace lattices

Let $R_d = E^0 B\Sigma_{p^d}$, and let $I^{\mathrm{tr}}_{d,\mathrm{univ}} \subset R_d$ be the transfer ideal from proper partition subgroups. Let $J^{\mathrm{pr}}_{d,\mathrm{univ}}$ be the intersection of the corresponding restriction kernels. Strickland's splitting theorem, read together with its correction, [41, 42, Theorem 1.1 and Sections 5–8] shows that these are direct summands as $\mathcal{O}$-modules and that multiplication by

$$c_d = e(V_{p^d} - 1)$$

induces an isomorphism

$$R_d / I^{\mathrm{tr}}_{d,\mathrm{univ}} \xrightarrow{\sim} J^{\mathrm{pr}}_{d,\mathrm{univ}}.$$

After fixed-source specialization put

$$A_d = R_d \widehat{\otimes}_{\mathcal{O},\epsilon_W} W(k),$$
$$\overline{I}^{\mathrm{tr}}_d = I^{\mathrm{tr}}_{d,\mathrm{univ}} \widehat{\otimes}_{\mathcal{O}} W(k),$$
$$\overline{J}^{\mathrm{pr}}_d = J^{\mathrm{pr}}_{d,\mathrm{univ}} \widehat{\otimes}_{\mathcal{O}} W(k).$$

Lemma 5.1 (Source specialization of the transfer splitting)**.** *The quotient and primitive summand remain split finite projective modules after fixed-source specialization, and multiplication by c_d induces*

$$B_d = A_d / \overline{I}^{\mathrm{tr}}_d \xrightarrow{\sim} \overline{J}^{\mathrm{pr}}_d.$$

Moreover,

$$\mathrm{Ann}_{A_d}(\overline{I}^{\mathrm{tr}}_d) = \overline{J}^{\mathrm{pr}}_d,$$

and the source-specialized $K(h)$-local Frobenius pairing on A_d is perfect.

Proof. Before specialization the transfer ideal and the primitive module are individually split finite-projective $\mathcal{O}$-submodules, and multiplication by c_d identifies the transfer quotient with the primitive module [41, Theorem 8.6]. They are not asserted to be complementary. The two split inclusions and this isomorphism survive source specialization, even though the section need not be flat. This gives the displayed isomorphism over $W(k)$.

To handle the annihilator without appealing to commutation of annihilators with nonflat base change, let $\varepsilon_d^{\mathrm{univ}} : R_d \to \mathcal{O}$ be the universal $K(h)$-local transfer and consider

$$\Phi_d^{\mathrm{univ}} : R_d \longrightarrow \bigl(I_{d,\mathrm{univ}}^{\mathrm{tr}}\bigr)^\vee, \qquad a \longmapsto \bigl(i \mapsto \varepsilon_d^{\mathrm{univ}}(ai)\bigr).$$

The perfect Frobenius pairing identifies R_d with $R_d^\vee$. Since $I_{d,\mathrm{univ}}^{\mathrm{tr}} \hookrightarrow R_d$ is a split inclusion of finite projective modules, dual restriction is split surjective. Hence

$$0 \longrightarrow J_{d,\mathrm{univ}}^{\mathrm{pr}} \longrightarrow R_d \xrightarrow{\Phi_d^{\mathrm{univ}}} \bigl(I_{d,\mathrm{univ}}^{\mathrm{tr}}\bigr)^\vee \longrightarrow 0$$

is split exact. Base change preserves this exact sequence and shows that $\overline{J}_d^{\mathrm{pr}}$ is the kernel of the specialized map

$$\Phi_d : A_d \longrightarrow \bigl(\overline{I}_d^{\mathrm{tr}}\bigr)^\vee.$$

The specialized pairing remains perfect. If $\Phi_d(a) = 0$, then $\varepsilon_d(air) = 0$ for every $i \in \overline{I}_d^{\mathrm{tr}}$ and $r \in A_d$, because the transfer ideal is an ideal. Perfection forces $ai = 0$ for every i, so $\ker(\Phi_d) = \mathrm{Ann}_{A_d}(\overline{I}_d^{\mathrm{tr}})$. □

Write

$$P_d = \mathrm{Ann}_{A_d}(\overline{I}_d^{\mathrm{tr}}) = \overline{J}_d^{\mathrm{pr}} = c_d B_d.$$

Let $\varepsilon_d : A_d \to W(k)$ be the $K(h)$-local transfer along $B\Sigma_{p^d} \to *$. By lemma 5.1 and Strickland's duality theorem, the resulting pairing is Frobenius and perfect, compatibly with generalized characters [40, 17].

Lemma 5.2 (Primitive–indecomposable duality)**.** *The pairing*

$$P_d \times B_d \longrightarrow W(k), \qquad (x, \bar{y}) \longmapsto \varepsilon_d(xy)$$

is well-defined and perfect. Thus it induces an isomorphism

$$\Psi_d : P_d \xrightarrow{\sim} B_d^\vee.$$

Proof. The product of P_d with the transfer ideal vanishes. In a Frobenius algebra the orthogonal complement of an ideal is its annihilator: if $\varepsilon_d(air) = 0$ for every i in the ideal and every r, perfection forces $ai = 0$. The assertion follows from the identification of the annihilator with P_d. □

Let $q_d : P_d \to B_d$ be induced by $A_d \to B_d$, and let

$$\lambda_d : B_d \longrightarrow B_d^\vee, \qquad \lambda_d(b)(b') = \mathrm{Tr}_{B_d/W(k)}(bb')$$

be the ordinary trace-dual map.

Lemma 5.3 (Character normalization of the two pairings)**.** *Let C be a faithfully flat HKR character extension of $W(k)[1/p]$ after source specialization, and let*

$$\mathscr{S}_{h,d} = \{H \leq \mathbf{Z}_p^h : [\mathbf{Z}_p^h : H] = p^d\}.$$

There is a character isomorphism

$$B_d \otimes C \xrightarrow{\sim} \prod_{H \in \mathscr{S}_{h,d}} C. \tag{5.1}$$

For $a \in P_d$, $b \in B_d$, write a_H, b_H for their components. Then

$$(\Psi_d(a) \otimes 1)(b) = \varepsilon_d(ab) = p^{-d} \sum_{H \in \mathscr{S}_{h,d}} a_H b_H, \tag{5.2}$$

$$((\lambda_d q_d)(a) \otimes 1)(b) = \sum_{H \in \mathscr{S}_{h,d}} a_H b_H. \tag{5.3}$$

The comparison is natural under source specialization and further faithfully flat extension of the character ring.

Proof. The character isomorphism (5.1) is the transitive part of the HKR description of the transfer quotient. A primitive class vanishes on every nontransitive component. The ordinary algebra trace of the split product is the unweighted component sum, which gives (5.3).

For (5.2), the HKR–Ganter transfer formula groups homomorphisms $\mathbf{Z}_p^h \to \Sigma_{p^d}$ by conjugacy class and weights each class by the inverse order of its centralizer [17, 12, 40]. The transitive action indexed by H factors through the left regular action of $A_H = \mathbf{Z}_p^h/H$. The centralizer of the left regular subgroup in $\mathrm{Sym}(A_H)$ is the right regular subgroup: a commuting permutation is determined by the image of the identity and is right translation by that element. Hence its order is $|A_H| = p^d$, giving the factor p^{-d}.

The character map, transfer, ordinary trace, and source slice are all compatible with scalar extension. The two formulas therefore remain valid after any further faithfully flat character extension. □

Theorem 5.4 (Exact Euler–trace comparison)**.** *The diagram*

$$\begin{array}{ccc} P_d & \xrightarrow{q_d} & B_d \\ {\scriptstyle \Psi_d}\downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \lambda_d} \\ B_d^\vee & \xrightarrow{p^d} & B_d^\vee \end{array}$$

commutes. Equivalently, as maps $P_d \to B_d^\vee$,

$$\lambda_d q_d = p^d \Psi_d.$$

Proof. The two character-side formulas in lemma 5.3 differ by the scalar p^d, so the diagram commutes after the faithfully flat character extension over $W(k)[1/p]$. Both composites are maps between finite free $W(k)$-modules. Their difference vanishes after inverting p, and the target is p-torsion-free;

the equality therefore descends integrally. The possible Witt–Frobenius twist on the coefficient copy of $W(k)$ acts on both sides through the same automorphism and does not alter the comparison. □

Put

$$\Theta_d = q_d \Psi_d^{-1} : B_d^\vee \longrightarrow B_d.$$

Then

$$\lambda_d \Theta_d = p^d \, \mathrm{id}_{B_d^\vee}.$$

In particular every Smith exponent of the trace pairing is at most d.

Lemma 5.5 (Primitive Euler component)**.** *Fix Strickland's complex orientation and its associated coordinate x. After a generic level-structure splitting, the component of the image $\bar{c}_d \in B_d$ indexed by an order-p^d subgroup D is*

$$\bar{c}_d(D) = \prod_{0 \neq a \in D} x(a).$$

A change of orientation or dual-character convention multiplies each component by a unit and does not change any valuation below.

Proof. A transitive abelian p-subgroup acts through its regular representation, whose reduced representation is the direct sum of its nontrivial characters. Under the chosen complex orientation, the Euler class of a character is the coordinate of the associated level-structure point. The tautological level structure identifies the nontrivial characters with the nonzero points of D. □

Lemma 5.6 (Exact-order torsion product)**.** *For every $s \geq 1$,*

$$v_p \left(\prod_{\mathrm{ord}(a) = p^s} x(a) \right) = 1.$$

Proof. Over a valued splitting extension, Weierstrass factorization for the formal multiplication series [22, 14] gives

$$[p^s](X) = U_s(X) X \prod_{0 \neq a \in F[p^s]} (X - x(a)), \qquad U_s(0) \in W(k)^\times.$$

Differentiating at the origin and using $[p^s]'(0) = p^s$ shows that the product over all nonzero p^s-torsion points has valuation s. Subtract the corresponding statement for p^{s-1}. □

Proposition 5.7 (Euler norm valuation)**.** *One has*

$$v_p \, \mathrm{Norm}_{B_d / W(k)}(\bar{c}_d) = L_d.$$

Proof. After splitting B_d, its norm is the product of the components from lemma 5.5. A fixed point of exact order p^s belongs to exactly $N_{h,d-s}$ subgroups of order p^d: after quotienting by the cyclic subgroup it generates,

the ambient divisible group is again isomorphic to $(\mathbf{Q}_p/\mathbf{Z}_p)^h$. Grouping by exact order and applying lemma 5.6 gives

$$v_p \operatorname{Norm}(\bar{c}_d) = \sum_{s=1}^{d} N_{h,d-s} = L_d.$$

□

Theorem 5.8 (Trace-lattice colength)**.** *For every finite height and every $d \geq 0$,*

$$\operatorname{length}_{W(k)}(B_d/\mathcal{T}^d(B_d)) = L_d.$$

Proof. Let $a_{d,1}, \ldots, a_{d,N_{h,d}}$ be the Smith exponents of λ_d. The identity $\lambda_d \Theta_d = p^d$ gives $0 \leq a_{d,i} \leq d$. Under $P_d \simeq c_d B_d$, the determinant of Θ_d is, up to a unit, the norm of $\bar{c}_d$. Hence

$$v_p(\det \lambda_d) = dN_{h,d} - L_d.$$

In Smith coordinates, membership in $\lambda_d^{-1}(p^d B_d^\vee)$ requires the ith coordinate to be divisible by $p^{d-a_{d,i}}$. Its quotient colength is therefore

$$\sum_i (d - a_{d,i}) = dN_{h,d} - v_p(\det \lambda_d) = L_d.$$

□

Corollary 5.9 (Discriminant valuation of the endpoint algebra)**.** *The trace discriminant of $B_d/W(k)$ satisfies*

$$v_p \operatorname{Disc}(B_d/W(k)) = dN_{h,d} - L_d.$$

Proof. The discriminant is the determinant of the trace-dual map λ_d. The determinant computation in the proof of theorem 5.8 gives the formula. □

Proposition 5.10 (The first trace layer)**.** *Let $n_1 = N_{h,1} = 1 + p + \cdots + p^{h-1}$. The Smith exponents of the trace pairing on B_1 are*

$$0, \underbrace{1, \ldots, 1}_{n_1 - 1 \text{ times}}.$$

In particular, $B_1/\mathcal{T}^1(B_1)$ has length one.

Proof. The identity $\lambda_1 \Theta_1 = p$ shows that every Smith exponent is either zero or one. The special fiber B_1/pB_1 is a local Artin algebra of k-dimension n_1. Multiplication by a nilpotent element has trace zero, whereas a scalar acts with trace n_1 times that scalar. Since $n_1 \equiv 1 \pmod{p}$, the reduced trace pairing has rank one. Exactly one Smith exponent is zero and the remaining $n_1 - 1$ exponents are one. The trace lattice at depth one therefore has colength $1 \cdot n_1 - (n_1 - 1) = 1$. □

6. The all-height geometric Witt filtration

Define

$$J_0 = \mathcal{O}, \qquad J_{d+1} = pJ_d + E_d.$$

Whenever $E_d \subseteq J_d$, put

$$K_d = J_d/E_d \subseteq B_d.$$

Lemma 6.1 (Saturation without a presentation)**.** *If $E_d \subseteq J_d$, then*

$$J_{d+1} \cap p\mathcal{O} = pJ_d.$$

Multiplication by p gives a short exact sequence

$$0 \longrightarrow \mathcal{O}/J_d \xrightarrow{p} \mathcal{O}/J_{d+1} \longrightarrow \mathcal{O}/(p, E_d) \longrightarrow 0.$$

Proof. Write $z = pj + e \in J_{d+1} \cap p\mathcal{O}$, with $j \in J_d$ and $e \in E_d$. The endpoint ideal is p-saturated, so $e = pe'$ with $e' \in E_d \subseteq J_d$. Thus $z = p(j + e')$. □

Lemma 6.2 (Geometric quotient inside the trace lattice)**.** *If $E_s \subseteq J_s$ for $s \leq d$, then*

$$K_d \subseteq \mathcal{T}^d(B_d).$$

Proof. Expanding the recursion gives

$$J_d = p^d\mathcal{O} + \sum_{s=0}^{d-1} p^{d-1-s}E_s.$$

The image of the first summand lies in p^dB_d. For $x \in E_s$, $s < d$, theorem 4.7 gives $\tau_d(x) \in \mathcal{T}^{s+1}(B_d)$, and multiplication by p^{d-1-s} places it in $\mathcal{T}^d(B_d)$. □

Theorem 6.3 (All-depth endpoint carry)**.** *For every $d \geq 0$,*

$$E_d \subseteq J_d, \qquad K_d = \mathcal{T}^d(B_d), \qquad \operatorname{length}_{W(k)}(\mathcal{O}/J_d) = L_d.$$

Proof. Induct on d. At $d = 0$, one has $B_0 = W(k)$, $E_0 = (u_1, \ldots, u_{h-1})$, and $K_0 = B_0 = \mathcal{T}^0(B_0)$. Assume the result through d. Repeated use of lemma 6.1 and

$$\operatorname{length}_k \mathcal{O}/(p, E_s) = \operatorname{rank}_{W(k)} B_s = N_{h,s}$$

gives $\operatorname{length}(\mathcal{O}/J_d) = L_d$. Hence

$$\operatorname{length}(B_d/K_d) = L_d.$$

The preceding lemma and theorem 5.8 give an inclusion of sublattices with equal colength, so $K_d = \mathcal{T}^d(B_d)$.

Now theorem 4.7 places $\beta_{d+1}(E_{d+1})$ in K_d. Given $x \in E_{d+1}$, choose $j \in J_d$ mapping to $\beta_{d+1}(x)$. Then $\tau_d(x - pj) = 0$, so $x - pj \in E_d$ and $x \in E_d + pJ_d = J_{d+1}$. The next length formula follows from lemma 6.1. □

6.1. **Integral recognition from degree-one operations.** Let

$$\mathbb{P} = \bigoplus_{m\geq 0} \mathbb{P}_m$$

denote Rezk's weight-graded algebraic power-operation monad on $\mathcal{O}$-modules. For a finite free module M, the $\mathfrak{m}$-adic completion of $\mathbb{P}(M)$ computes the degree-zero homotopy of the corresponding free $K(h)$-local commutative E-algebra; see [29, Section 1.3] and [31, Sections 3.6 and 3.14].

Let $M_X = \mathcal{O}\{X\}$ be free of rank one in even degree and put

$$\mathscr{F}_X = \bigoplus_{m\geq 0} \mathbb{P}_m(M_X).$$

The map $X \mapsto 0$ gives its augmentation $\mathscr{F}_X \to \mathcal{O}$, with augmentation ideal $(\mathscr{F}_X)_+$. When a complete target is used, the free object is the $\mathfrak{m}$-adic completion

$$\widehat{\mathscr{F}_X} = \varprojlim_j \mathscr{F}_X/\mathfrak{m}^j\mathscr{F}_X.$$

Every weight summand is finite free over the complete ring $\mathcal{O}$, so completion does not change any fixed-weight summand or its indecomposable quotient.

Let

$$\mathscr{C}_E = \bigoplus_{m\geq 0} E_0^\vee B\Sigma_m, \qquad (\mathscr{C}_E)_m = E_0^\vee B\Sigma_m.$$

For $i, j \geq 0$, multiplication in weights i and j is the composite

$$\begin{aligned}\mu_{i,j} : E_0^\vee B\Sigma_i \otimes_{\mathcal{O}} E_0^\vee B\Sigma_j &\xrightarrow[\cong]{\boxtimes} E_0^\vee B(\Sigma_i \times \Sigma_j) \\ &\xrightarrow{\mathrm{tr}_{i,j}} E_0^\vee B\Sigma_{i+j},\end{aligned} \tag{6.1}$$

where $\mathrm{tr}_{i,j}$ denotes the covariant block-sum product in completed E-homology, dual under the finite-free pairing to cohomological restriction along $\Sigma_i \times \Sigma_j \subseteq \Sigma_{i+j}$. This specifies the variance of the transfer-product notation used here. The summand $E_0^\vee B\Sigma_0 = \mathcal{O}$ is the unit, and projection to weight zero is the augmentation; compare [33, Section 3]. Rezk's corepresentability theorem identifies $\mathscr{C}_E$ with the graded rank-one free object:

$$\iota_X : \mathscr{F}_X \xrightarrow{\sim} \mathscr{C}_E, \qquad \mathbb{P}_m(M_X) \xrightarrow{\sim} E_0^\vee B\Sigma_m \quad (m \geq 0). \tag{6.2}$$

See [33, Proposition 3.1 and Lemma 3.2]; the operation-side description is [31, Sections 3.6 and 3.14].

Write

$$Q(\mathscr{F}_X) = (\mathscr{F}_X)_+/(\mathscr{F}_X)_+^2.$$

Let

$$\Delta = \bigoplus_{e\geq 0} \Delta[e]$$

be Rezk's graded ring of natural operations on the indecomposable, or cotangent, functor of augmented $\mathbb{P}$-algebras. Each $\Delta[e]$ is an $\mathcal{O}$-bimodule. The left action multiplies the output of an operation, while the right action

inserts a coefficient before the operation. Composition is therefore the balanced product

$$\Delta[e] \otimes_{\mathcal{O}} \Delta[f] \longrightarrow \Delta[e+f],$$

using the right action on the first factor and the left action on the second.

Evaluation at the universal generator gives the canonical isomorphism

$$\mathrm{ev}_X : \Delta[e] \xrightarrow{\sim} Q(\mathscr{F}_X)_{p^e}, \qquad D \longmapsto [D(X)], \tag{6.3}$$

of [31, Proposition 3.15]. The comparison between the additive-operation ring and Δ is [31, Corollary 3.20].

Let $\Gamma[1]$ be the finite free module of additive order-p operations, and choose a representative $\sigma \in \Gamma[1]$ of Rezk's Frobenius class.

Lemma 6.4 (The decomposable line in weight p)**.** *The decomposable weight-p submodule of $\mathscr{F}_X$ is the single line*

$$((\mathscr{F}_X)^2_+)_p = \mathcal{O}\{X^p\}.$$

Proof. Every proper decomposition of the weight p uses regular weights $1, \dots, p-1$. In those weights Rezk's regular-weight comparison identifies the rank-one free power-operation algebra with the ordinary symmetric algebra; see [30, Sections 9–10, especially Proposition 10.9]. Every product of positive-weight elements of total weight p is therefore a multiple of X^p. Conversely X^p is decomposable, proving the equality. □

Proposition 6.5 (The weight-p pushout and indecomposables)**.** *Let r denote the generator of a free rank-one copy of $\mathcal{O}$. Rezk's weight-p construction is the pushout*

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{a \mapsto pa\, r} & \mathcal{O}\{r\} \\ {\scriptstyle a \mapsto (a\sigma, -aX^p)}\downarrow & & \downarrow{\scriptstyle r \mapsto \Theta(X)} \\ \Gamma[1] \oplus \mathcal{O}\{X^p\} & \xrightarrow{(Q, bX^p) \mapsto Q(X) + bX^p} & (\mathscr{F}_X)_p, \end{array} \tag{6.4}$$

where $p\Theta(X) = \sigma(X) - X^p$. Let $\delta \in \Delta[1]$ be the element corresponding under (6.3) *to $[\Theta(X)] \in Q(\mathscr{F}_X)_p$; equivalently, δ is obtained from the upper-right generator r by the right vertical map in* (6.4) *followed by passage to indecomposables. Then*

$$0 \longrightarrow \mathcal{O} \xrightarrow{a \mapsto (a\sigma, -pa)} \Gamma[1] \oplus \mathcal{O}\delta \longrightarrow \Delta[1] \longrightarrow 0. \tag{6.5}$$

Writing $n_1 = N_{h,1}$, one may choose an $\mathcal{O}$-basis $\sigma, Q_1, \dots, Q_{n_1-1}$ of $\Gamma[1]$ such that

$$\delta, q_1, \dots, q_{n_1-1}, \qquad q_i = [Q_i(X)],$$

is an $\mathcal{O}$-basis of $\Delta[1]$.

Proof. The pushout is Rezk's weight-p integrality square [30, Proposition 10.6]; the two paths from the upper-left corner express the relation $\sigma(X) - X^p = p\Theta(X)$. By lemma 6.4, quotienting the pushout by decomposables kills exactly the line generated by X^p and leaves the relation $\sigma = p\delta$, which is exactly (6.5). The left map is injective because $\mathcal{O}$ is p-torsion-free.

Both $\Gamma[1]$ and $\Delta[1]$ are finite free of rank n_1. Since the cokernel is projective, the relation vector $(\sigma, -p)$ is unimodular. Modulo the maximal ideal its second component vanishes, so σ is primitive and extends to a basis; the displayed basis of $\Delta[1]$ follows. □

Put $\mathcal{D} = \{\Theta, Q_1, \dots, Q_{n_1-1}\}$. Words are composed rightmost first, and coefficients are written on the left after applying the balanced bimodule relations.

Lemma 6.6 (Coefficient transport and word bases)**.** *For every $e \geq 0$, pure length-e words in $\mathcal{D}$ span $\Delta[e]$ as a left $\mathcal{O}$-module. There is a finite subset $\mathcal{W}_e$ whose classes form an $\mathcal{O}$-basis of $\Delta[e]$.*

Proof. Let $D_0 = \delta, D_1 = q_1, \dots, D_{n_1-1} = q_{n_1-1}$ be the left basis of $\Delta[1]$. For $a \in \mathcal{O}$, write its right action in this basis as

$$D_i a = \sum_j c_{ji}(a) D_j, \qquad c_{ji}(a) \in \mathcal{O}.$$

No centrality is assumed. In a balanced tensor product,

$$\begin{aligned}(a_1 D_i) \otimes (a_2 D_k) &= a_1 (D_i a_2) \otimes D_k \\ &= \sum_j a_1 c_{ji}(a_2) D_j \otimes D_k.\end{aligned}$$

For a longer tensor, expand every factor in the left basis, start at the last factor, and move its coefficient into the right action on the preceding factor. Re-expand that factor and continue towards the first one. This terminates and puts every coefficient on the far left.

Rezk's Koszul theorem gives degree-one generation of Δ [31, Theorem 1.7, Corollary 3.20, and Proposition 4.10]. Thus the balanced multiplication map $\Delta[1]^{\otimes_{\mathcal{O}} e} \to \Delta[e]$ is surjective, and pure words span. From this finite spanning family select a subset whose residues form a basis modulo the maximal ideal. Nakayama gives a surjection from a free module of rank $\operatorname{rank}_{\mathcal{O}} \Delta[e]$ onto $\Delta[e]$. Both modules are free of that rank, so the representing square matrix has unit determinant and is an isomorphism. At $e = 0$ the empty word is the identity operation. □

Lemma 6.7 (Composition of integral representatives)**.** *For a word $w = D_1 \cdots D_e$ in the chosen integral representatives, with the rightmost operation applied first, the class $[w(X)]$ in $Q(\mathscr{F}_X)$ is the evaluation at $[X]$ of the product of the corresponding elements of Δ.*

Proof. For an augmented $\mathbb{P}$-algebra A, write A_+ for its augmentation ideal. The construction in the proof of [31, Proposition 3.15] uses the functorial $\mathbb{P}$-algebra quotient $A \to A/A_+^2$. A universal representative $u(X) \in (\mathscr{F}_X)_+$ therefore induces on $Q(A) = A_+/A_+^2$ the operation

$$[z] \longmapsto [u(z)].$$

It is well-defined by naturality with respect to that quotient and depends only on $[u(X)]$. This applies to Θ as an integral universal operation: no division by p is being performed inside $Q(A)$. Successive application gives

$$[D_1(D_2(\cdots D_e(X)\cdots))] = D_1D_2\cdots D_e([X]).$$

The right coefficient action on the free-algebra indecomposables is substitution $X \mapsto aX$. It is balanced with the left action on the next operation, exactly as in lemma 6.6. □

Remark 6.8. The coefficient-transport identities are identities in Δ and its balanced tensor powers. They need not hold for the nonlinear representatives on the underlying ring. Only their indecomposable classes are used when selecting word bases.

Lemma 6.9 (Indecomposable identification)**.** *The composite*

$$\Delta[e] \xrightarrow{\mathrm{ev}_X} Q(\mathscr{F}_X)_{p^e} \xrightarrow{Q(\iota_X)} Q(\mathscr{C}_E)_{p^e}$$

is an $\mathcal{O}$-bimodule isomorphism. For $w \in \mathcal{W}_e$, put

$$Y_w = \iota_X(w(X)) \in (\mathscr{C}_E)_{p^e}.$$

Then the classes of the Y_w form an $\mathcal{O}$-basis of $Q(\mathscr{C}_E)_{p^e}$.

Proof. The first arrow is Rezk's evaluation isomorphism [31, Proposition 3.15], with the right action given by substitution $X \mapsto aX$. The second arrow is induced by the weightwise algebra isomorphism (6.2) and respects the same substitution. By lemma 6.7, $[w(X)]$ evaluates the product represented by the word w. The word basis of lemma 6.6 therefore maps to the asserted basis of $Q(\mathscr{C}_E)_{p^e}$. □

Lemma 6.10 (Polynomial lifting from indecomposables)**.** *Let A be a graded polynomial $\mathcal{O}$-algebra, finite free in each graded piece. Suppose homogeneous elements y_λ lift an $\mathcal{O}$-basis of $Q(A) = A_+/A_+^2$ in every degree. Then the map from the polynomial algebra on the y_λ to A is an isomorphism.*

Proof. Induction on the degree shows surjectivity: after subtracting a lift of the required indecomposable class, the remainder is decomposable and is a sum of products of lower-degree elements. The source and target have the same Hilbert series because their indecomposable generators occur with the same degrees and multiplicities. A graded surjection between finite free graded pieces of equal rank is an isomorphism in every degree. □

Theorem 6.11 (General-height amplified coordinates)**.** *The family $\{Y_w : w \in \mathcal{W}_e,\ e \geq 0\}$ is a homogeneous polynomial coordinate system for $\mathscr{C}_E$. If $c(x) : \mathscr{C}_E \to k$ is the map corresponding to $x \in \mathcal{O}$ under cofreeness, then*

$$c(x)(Y_w) = \epsilon_k(w(x)). \tag{6.6}$$

Thus

$$x \in I_d \quad \Longleftrightarrow \quad \epsilon_k(w(x)) = 0 \text{ for every } w \in \mathcal{W}_e \text{ with } e < d.$$

Proof. Weight m of the rank-one free object is canonically $E_0^\vee B\Sigma_m$. Under these identifications, multiplication agrees with (6.1): both use the block-sum extended-power construction, equivalently the dual of cohomological restriction. Thus (6.2) is an algebra isomorphism, not merely a comparison of ranks. By lemma 6.9, the classes $[Y_w]$ give a basis of the indecomposables in each weight p^e. Polynomiality of $\mathscr{C}_E$ [33, Lemma 3.2] and lemma 6.10 imply that the Y_w are polynomial coordinates.

For $x \in \mathcal{O}$, the universal property gives an evaluation map of power-operation algebras

$$\mathrm{ev}_x : \mathscr{F}_X \longrightarrow \mathcal{O}, \qquad X \longmapsto x.$$

Its weight-m restriction, paired against $E^0 B\Sigma_m$, is evaluation against the total power operation $P_m(x)$; this is the defining extended-power construction of P_m. Hence

$$\epsilon_k \circ \mathrm{ev}_x \circ \iota_X^{-1} : \mathscr{C}_E \longrightarrow k$$

corepresents precisely the Witt vector obtained by applying the total power operation to x and reducing coefficients by ϵ_k. This is the canonical map $\mathcal{O} \to \mathbb{W}_E(k)$ that cofreeness identifies as an isomorphism [4, 29]. It is therefore the point $c(x)$. Equivalently, the following square commutes:

$$\begin{array}{ccc} \mathscr{F}_X & \xrightarrow{\mathrm{ev}_x} & \mathcal{O} \\ {\scriptstyle \iota_X}\big\downarrow{\scriptstyle \sim} & & \big\downarrow{\scriptstyle \epsilon_k} \\ \mathscr{C}_E & \xrightarrow{c(x)} & k. \end{array}$$

Naturality of evaluation sends $w(X)$ to $w(x)$ and proves (6.6).

The zero Witt vector is zero on every positive-weight piece of $\mathscr{C}_E$. By the definition of the filtration [33, Section 4], $x \in I_d$ exactly when $c(x)$ vanishes on weights $0 < m < p^d$. In a positively graded polynomial algebra this is equivalent to vanishing on the polynomial generators in those weights. Their weights are exactly p^e, yielding the asserted criterion.

All constructions are weightwise. Each fixed-weight piece is finite free over the complete ring $\mathcal{O}$, so completion does not change it. The continuous extension to $\widehat{\mathscr{F}_X}$ gives the same formulas. Evaluation at a general x need not preserve the augmentation $X \mapsto 0$; that augmentation is used only to form indecomposables. □

Corollary 6.12 (One-step recognition)**.** *Let $d \geq 1$ and $x \in I_d$. If $D(x) \in I_d$ for every $D \in \mathcal{D}$, then $x \in I_{d+1}$.*

Proof. All coordinates of length less than d vanish because $x \in I_d$. For $w = D_1 \cdots D_d \in \mathcal{W}_d$, put $y = D_d(x) \in I_d$. The element $D_1 \cdots D_{d-1}(X)$ of the free algebra has positive weight $p^{d-1} < p^d$, whether or not this shorter word was selected for a word basis. The defining filtration condition on y therefore gives $\epsilon_k(D_1 \cdots D_{d-1}(y)) = 0$. This is $\epsilon_k(w(x)) = 0$. All length-d coordinates vanish, and theorem 6.11 gives $x \in I_{d+1}$. □

6.2. **Flag divisibility and completion of the proof.** The operation-theoretic input now meets the quotient geometry. Let $s_1, t_1 : \mathcal{O} \to \mathcal{O}_1$ be the source and target maps of the universal order-p subgroup ring. Define the two-step flag algebra

$$\mathcal{F}_d = B_{d-1} \widehat{\otimes}_{\mathcal{O}, \tau_{d-1}, s_1} \mathcal{O}_1. \tag{6.7}$$

For a complete local test algebra, it classifies an order-p^{d-1} subgroup D of the fixed source lift F, followed by an order-p subgroup D' of F/D. The inverse image $\widetilde{D} \subset F$ has order p^d, and the natural isomorphism $(F/D)/D' \simeq F/\widetilde{D}$ respects the induced marking of the special fiber. These constructions are functorial in the test algebra. Yoneda therefore gives a map

$$\iota_d : B_d \longrightarrow \mathcal{F}_d.$$

Let

$$\psi_{d-1} : \mathcal{O} \longrightarrow \mathcal{F}_d, \qquad x \longmapsto 1 \otimes t_1(x)$$

be the base-changed total order-p operation. Compatibility of final targets is the commutative square

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{\tau_d} & B_d \\ {\scriptstyle \psi_{d-1}}\downarrow & & \downarrow{\scriptstyle \iota_d} \\ \mathcal{F}_d & = & \mathcal{F}_d. \end{array} \tag{6.8}$$

Regard $\mathcal{O}_1$ as a finite free left $\mathcal{O}$-module through s_1. Finite-free duality identifies

$$\Gamma[1] \simeq \mathrm{Hom}_{\mathcal{O}}({}_{s_1}\mathcal{O}_1, \mathcal{O}). \tag{6.9}$$

For $Q \in \Gamma[1]$, let ℓ_Q be the corresponding functional and extend it to

$$\ell_{Q,d} : \mathcal{F}_d \longrightarrow B_{d-1}, \qquad \ell_{Q,d}(b \otimes u) = b\, \tau_{d-1}(\ell_Q(u)).$$

For $a \in \mathcal{O}$, source-linearity gives $\ell_Q(s_1(a)u) = a\ell_Q(u)$, so

$$b\, \tau_{d-1}(a)\tau_{d-1}(\ell_Q(u)) = b\, \tau_{d-1}(\ell_Q(s_1(a)u)).$$

Thus the two representatives of the balancing relation have the same image, proving that $\ell_{Q,d}$ is well-defined. The coefficient functional description of the total operation gives

$$\ell_{Q,d}(1 \otimes t_1(x)) = \tau_{d-1}(Q(x)). \tag{6.10}$$

Proposition 6.13 (Scheme-theoretic flag divisibility)**.** *For every additive degree-one operation $Q \in \Gamma[1]$ and every $d \geq 1$,*

$$Q(E_d) \subseteq E_{d-1}.$$

Proof. If $x \in E_d$, then $\tau_d(x) = 0$. The commutative square (6.8) gives $\psi_{d-1}(x) = 1 \otimes t_1(x) = 0$ in $\mathcal{F}_d$. Applying the arbitrary coefficient functional $\ell_{Q,d}$ and using (6.10) yields $\tau_{d-1}(Q(x)) = 0$. Hence $Q(x) \in E_{d-1}$. □

For $x \in E_d$, the Bockstein satisfies $\tau_{d-1}(x) = p\beta_d(x)$. Applying the Frobenius-class relation and proposition 6.13 gives the following.

Lemma 6.14 (Divided Frobenius from the endpoint Bockstein)**.** *For $x \in E_d$,*

$$\tau_{d-1}(\Theta x) = -p^{p-1}\beta_d(x)^p.$$

If $\beta_d(x) \in K_{d-1}$, then $\Theta x \in J_d$.

Proof. The class $\sigma(x)$ lies in E_{d-1}. Apply τ_{d-1} to

$$\sigma(x) = x^p + p\Theta(x)$$

and use $\tau_{d-1}(x) = p\beta_d(x)$. Since B_{d-1} is p-torsion-free, cancellation gives the formula. If $\beta_d(x) \in K_{d-1}$, the right side lies in pK_{d-1}. The inverse image of pK_{d-1} under τ_{d-1} is $E_{d-1} + pJ_{d-1} = J_d$. □

Proof of theorem 4.2. The base case follows from the characteristic-p endpoint identity and Rezk's saturation theorem [33, Theorem 7.1]:

$$I_1 = (p, u_1, \dots, u_{h-1}) = p\mathcal{O} + E_0 = J_1.$$

Assume $I_d = J_d$, with $d \geq 1$, and let $x \in E_d$. The endpoint carry theorem gives $x \in J_d = I_d$. Every complementary additive basis operation sends x into $E_{d-1} \subseteq J_d = I_d$ by proposition 6.13. Also

$$\beta_d(x) \in \mathcal{T}^{d-1}(B_{d-1}) = K_{d-1},$$

so the divided-Frobenius lemma gives $\Theta(x) \in J_d = I_d$. It follows that every operation in the integral degree-one basis sends x into I_d, and corollary 6.12 gives $E_d \subseteq I_{d+1}$. Rezk's saturation identity [33, Theorem 7.1] gives $pI_d \subseteq I_{d+1}$, so

$$J_{d+1} = pI_d + E_d \subseteq I_{d+1}.$$

By theorem 6.3 and Rezk's associated-graded rank and length formula [33, Proposition 4.4 and Section 8], the two quotient rings have the same finite $W(k)$-length L_{d+1}. The inclusion is therefore an equality. □

Remark 6.15 (Composition and coefficient actions). All traces in the proof use the source coefficient structure, with target Frobenius actions retained as in remark 4.1. In particular, no depthwise untwisting is applied to the ghost sequence. Operations are composed rightmost first, both in the flag algebra and in the cofreeness evaluation. The latter ends with ϵ_k, not ϵ_W.

7. Mixed-characteristic layers and invariant factors

We use the elementary-divisor classification for finite modules over the DVR $W(k)$; see [8, Chapter 6]. Let $\mathcal{O}_h$ be the height-h Lubin–Tate deformation ring and let $\{I_r^{(h)}\}$ be Rezk's Witt filtration. Put $A_{h,r} = \mathcal{O}_h/I_r^{(h)}$ and set $N_{h,-1} = 0$.

Theorem 7.1 (All-height p-adic layers)**.** *For $0 \leq s < r$, multiplication by p^s gives a canonical isomorphism*

$$p^s A_{h,r}/p^{s+1}A_{h,r} \simeq \mathcal{O}_h/(p, I_{r-s}^{(h)}).$$

It follows that

$$\dim_k(p^s A_{h,r}/p^{s+1}A_{h,r}) = N_{h,r-s-1}.$$

Proof. Rezk's multiplication theorem gives $pI_t^{(h)} = I_{t+1}^{(h)} \cap p\mathcal{O}_h$. Since $\mathcal{O}_h$ is p-torsion-free,

$$(I_{t+1}^{(h)} : p) = I_t^{(h)},$$

and iteration gives $(I_r^{(h)} : p^s) = I_{r-s}^{(h)}$. Multiplication by p^s therefore identifies $\mathcal{O}_h/I_{r-s}^{(h)}$ with $p^s A_{h,r}$; the next layer corresponds to the image of $p\mathcal{O}_h$. Rezk's characteristic-p subgroup theorem identifies the final quotient with the order-p^{r-s-1} target algebra, whose rank is $N_{h,r-s-1}$ by Strickland's subgroup count. □

Theorem 7.2 (All-height invariant factors)**.** *As a finite $W(k)$-module,*

$$A_{h,r} \simeq \bigoplus_{e=1}^{r} (W(k)/p^e)^{\oplus m_{h,r,e}}, \qquad m_{h,r,e} = N_{h,r-e} - N_{h,r-e-1}.$$

Equivalently, $m_{h,r,e}$ is the coefficient of T^{r-e} in

$$\prod_{j=1}^{h-1} (1 - p^j T)^{-1}.$$

Moreover

$$\sum_{r\geq 1} \operatorname{length}_{W(k)}(A_{h,r}) T^r = \frac{T}{(1-T)^2 \prod_{j=1}^{h-1}(1-p^j T)}.$$

Proof. For a finite module over the DVR $W(k)$, the dimension of $p^s A/p^{s+1}A$ is the number of invariant factors of exponent greater than s. By theorem 7.1, this number is $N_{h,r-s-1}$. Taking successive differences gives the displayed multiplicity. Multiplying the subgroup generating series by $1 - T$ gives

$$\sum_{n\geq 0} (N_{h,n} - N_{h,n-1}) T^n = \prod_{j=1}^{h-1} (1 - p^j T)^{-1}.$$

Finally, $\operatorname{length}(A_{h,r}) = \sum_{d=0}^{r-1} N_{h,d}$, and summing in r gives the last formula. □

Corollary 7.3 (Growth with the depth)**.** *Assume $h \geq 2$ and fix p. Put*

$$c_{h,p} = \prod_{i=1}^{h-1} (1 - p^{-i})^{-1}.$$

Then

$$N_{h,d} \sim c_{h,p} p^{(h-1)d}, \qquad \operatorname{length}_{W(k)}(A_{h,r}) \sim \frac{c_{h,p}}{p^{h-1}-1} p^{(h-1)r}.$$

At height one, $N_{1,d} = 1$ and $\operatorname{length}_{W(k)}(A_{1,r}) = r$ exactly.

Proof. For $h \geq 2$, use

$$N_{h,d} = p^{(h-1)d} \prod_{i=1}^{h-1} \frac{1 - p^{-(d+i)}}{1 - p^{-i}}$$

and sum the resulting geometric asymptotic from $d = 0$ to $r - 1$. The height-one formula follows directly from the classical filtration $I_r = p^r W(k)$. □

7.1. **Low-height forms of the invariant-factor law.** The formula specializes transparently in the first three heights. When $h = 1$, one has $N_{1,d} = 1$ for $d \geq 0$, so all successive differences vanish except the last one. Hence

$$A_{1,r} \simeq W(k)/p^r.$$

At height two,

$$N_{2,d} = 1 + p + \cdots + p^d, \qquad m_{2,r,e} = p^{r-e},$$

and therefore

$$A_{2,r} \simeq \bigoplus_{e=1}^{r} (W(k)/p^e)^{\oplus p^{r-e}}.$$

The height-two geometric theorem below upgrades this module-theoretic identity to an explicit intersection formula for two endpoint divisors.

For $h = 3$, the Gaussian-binomial difference is

$$N_{3,n} - N_{3,n-1} = p^n(1 + p + \cdots + p^n).$$

Thus

$$A_{3,r} \simeq \bigoplus_{e=1}^{r} (W(k)/p^e)^{\oplus p^{r-e}(1+p+\cdots+p^{r-e})}.$$

For example,

$$A_{3,2} \simeq W(k)/p^2 \oplus (W(k)/p)^{p+p^2},$$

while

$$A_{3,3} \simeq W(k)/p^3 \oplus (W(k)/p^2)^{p+p^2} \oplus (W(k)/p)^{p^2+p^3+p^4}.$$

These decompositions record the entire mixed-characteristic thickness, not only the total colength. In particular, any geometric presentation of the height-three endpoint ideals must reproduce every layer in this list.

Corollary 7.4 (Exponent and number of cyclic factors)**.** *The exponent of $A_{h,r}$ is p^r, and its minimal number of generators as a $W(k)$-module is $N_{h,r-1}$. More generally, the number of invariant factors of exponent at least p^e is $N_{h,r-e}$.*

Proof. These statements follow directly from theorem 7.2. The summand of exponent p^r occurs once, and the sum of the multiplicities with exponent at least p^e telescopes to $N_{h,r-e}$. □

8. Euler presentations, complete intersections, and trace recognition

The Euler–trace comparison contains more information than its determinant. Together with the endpoint carry it identifies two consecutive Witt quotients as quotients of the same one-dimensional ring by regular elements. This forces their minimal resolutions to have the same ranks. No complete-intersection hypothesis on an endpoint ideal is used here.

8.1. **Two consecutive quotients from one endpoint algebra.**

Theorem 8.1 (Euler presentation)**.** *For every $d \geq 0$, the Euler image $\bar{c}_d \in B_d$ is a non-zero-divisor and*

$$\mathcal{T}^d(B_d) = \bar{c}_d B_d. \tag{8.1}$$

For any lift $\widetilde{c}_d \in \mathcal{O}$ under τ_d, one has

$$\begin{aligned} I_d &= E_d + (\widetilde{c}_d), \qquad \mathcal{O}/I_d \simeq B_d/(\bar{c}_d), \\ I_{d+1} &= E_d + (p\widetilde{c}_d), \quad \mathcal{O}/I_{d+1} \simeq B_d/(p\bar{c}_d). \end{aligned} \tag{8.2}$$

These are isomorphisms of $\mathcal{O}$-algebras, where B_d has its target $\mathcal{O}$-action. Relative to the source copy of $W(k)$ they are φ^d-semilinear. For $d \geq 1$, $\bar{c}_d$ is a nonunit.

Proof. The generic fiber of $B_d/W(k)$ is étale, so the trace-dual map $\lambda_d : B_d \to B_d^\vee$ is injective. The exact comparison $\lambda_d q_d = p^d \Psi_d$ of theorem 5.4, with Ψ_d an isomorphism, gives

$$\lambda_d(q_d(P_d)) = p^d B_d^\vee.$$

Consequently $q_d(P_d) = \lambda_d^{-1}(p^d B_d^\vee)$: if $\lambda_d(b) = p^d\eta$, take $a = \Psi_d^{-1}(\eta)$ and use injectivity to obtain $b = q_d(a)$. The source-specialized Euler isomorphism $B_d \simeq P_d$ identifies q_d with multiplication by $\bar{c}_d$, which is injective. Hence (8.1) holds and $\bar{c}_d$ is regular.

By theorems 4.2 and 6.3, the image of I_d in B_d is $\mathcal{T}^d(B_d)$ and $E_d \subseteq I_d$. Taking inverse images gives the first line of (8.2); the recursion $I_{d+1} = pI_d + E_d$ gives the second. The semilinearity is precisely the coefficient convention of remark 4.1. Finally, $\operatorname{length}(\mathcal{O}/I_d) = L_d > 0$ for $d \geq 1$, so $\bar{c}_d$ cannot be a unit. At $d = 0$ the convention is $\bar{c}_0 = 1$. □

8.2. **Minimal-resolution rigidity.** For a finite module M over a Noetherian local ring $(R, \mathfrak{m}, k)$, write $\beta_i^R(M) = \dim_k \operatorname{Tor}_i^R(M, k)$ and $\mathcal{P}_M^R(t) = \sum_i \beta_i^R(M) t^i$. These are Betti numbers over the ambient ring R, not over the quotient ring.

Lemma 8.2 (A minimal cone for a regular element)**.** *Let B be a finite R-module and $a \in \mathfrak{m}$ a non-zero-divisor on B. Then*

$$\mathcal{P}_{B/aB}^R(t) = (1+t)\mathcal{P}_B^R(t).$$

Proof. Take a minimal free resolution $F_\bullet \to B$. Multiplication by a is a chain map $F_\bullet \to F_\bullet$. Its cone resolves B/aB, since the kernel of multiplication by a on B is zero. All entries of the cone differentials lie in $\mathfrak{m}$: the old differentials do by minimality, and the additional diagonal blocks are multiplication by a. Thus the cone is minimal. Its degree-i rank is $\beta_i^R(B) + \beta_{i-1}^R(B)$, proving the formula. □

Theorem 8.3 (Complete intersections at every height)**.** *For every finite height h, every $d \geq 0$ endpoint ideal E_d is generated by a regular sequence of*

length $h-1$. Every Witt ideal I_r, $r \geq 1$, is generated by a regular sequence of length h. More precisely,

$$\mathcal{P}^{\mathcal{O}}_{\mathcal{O}/I_r}(t) = (1+t)^h \quad (r \geq 1), \qquad \mathcal{P}^{\mathcal{O}}_{B_d}(t) = (1+t)^{h-1} \quad (d \geq 0). \tag{8.3}$$

In particular, the minimal number of endpoint generators is $h-1$, independently of depth, and all these quotients have Koszul minimal resolutions over $\mathcal{O}$.

Proof. Put $R = \mathcal{O}$, a regular local ring of dimension h. For $d \geq 1$, both $\widetilde{c}_d$ and $p\widetilde{c}_d$ lie in its maximal ideal. Their images in B_d are non-zero-divisors, because $\bar{c}_d$ is regular by theorem 8.1 and B_d is p-torsion-free. Apply lemma 8.2 to the two presentations in (8.2). It gives

$$\mathcal{P}^R_{R/I_d}(t) = (1+t)\mathcal{P}^R_{B_d}(t) = \mathcal{P}^R_{R/I_{d+1}}(t) \qquad (d \geq 1).$$

Now $I_1 = \mathfrak{m}_R$, so the Koszul resolution of the residue field has polynomial $(1+t)^h$. Induction gives the first formula in (8.3), and cancellation of $1+t$ gives the second for $d \geq 1$. At $d = 0$ the endpoint ideal is $(u_1, \ldots, u_{h-1})$, giving the same formula directly.

For a quotient R/J, its first Betti number is the minimal number of generators of J. Thus I_r has h generators and E_d has $h-1$. The quotients have dimensions zero and one, respectively, so these numbers equal the heights of the ideals. In the Cohen–Macaulay ring R, an ideal generated by its height many elements is generated by a regular sequence; see [8, Chapter 18]. Their Koszul resolutions are minimal and have the ranks just computed. □

The extra input beyond Gorensteinness is the rigidity relation (8.3). It determines the numbers of generators and the resolution ranks, though not canonical equations in the deformation parameters.

Corollary 8.4 (Compatible regular-sequence presentations)**.** *Choose a regular sequence $f_{d,1}, \ldots, f_{d,h-1}$ generating E_d. Then, for $d \geq 1$, the sequences*

$$(f_{d,1}, \ldots, f_{d,h-1}, \widetilde{c}_d), \qquad (f_{d,1}, \ldots, f_{d,h-1}, p\widetilde{c}_d)$$

are regular and generate I_d and I_{d+1}, respectively.

Proof. Append the indicated element to the endpoint regular sequence. It is regular on B_d by theorem 8.1; the ideal identities are (8.2). □

8.3. **A perfect trace linking form.**

Theorem 8.5 (Trace duality of a Witt truncation)**.** *Put $K_W = W(k)[1/p]$. The quotient $B_d/(\bar{c}_d)$ has the perfect, symmetric, associative $W(k)$-bilinear pairing*

$$(\bar{x}, \bar{y}) \longmapsto p^{-d} \operatorname{Tr}^{\text{source}}_{B_d/W(k)}(xy) \pmod{W(k)} \quad \text{in } K_W/W(k). \tag{8.4}$$

On $\mathcal{O}/I_d$ with its original coefficient action, the corresponding $W(k)$-bilinear form is

$$(\bar{x}, \bar{y}) \longmapsto \varphi^{-d}\Big(p^{-d} \operatorname{Tr}^{\text{source}}_{B_d/W(k)}(\tau_d(xy))\Big) \pmod{W(k)}.$$

Here the inverse Frobenius is a fixed-level transport of the pairing, not a modification of the ghost sequence.

Proof. The equality $\bar{c}_d B_d = \mathcal{T}^d(B_d)$ makes (8.4) well-defined in each argument. Its radical in B_d is exactly $\mathcal{T}^d(B_d)$ by definition. It therefore embeds the finite-length quotient in its $W(k)$-dual with values in $K_W/W(k)$. That dual has the same length, so the embedding is an isomorphism. Commutativity gives symmetry and associativity. Applying φ^{-d} compensates exactly for the coefficient action of τ_d and proves the last assertion. □

8.4. **The complete trace Smith spectrum.** Put $m_{h,a} = N_{h,a} - N_{h,a-1}$, with $N_{h,-1} = 0$.

Theorem 8.6 (All-depth trace Smith factors)**.** *The trace-dual map $\lambda_d : B_d \to B_d^\vee$ has Smith exponent a with multiplicity $m_{h,a}$, for each $0 \le a \le d$. Equivalently, there are source $W(k)$-bases in which*

$$\lambda_d \sim \bigoplus_{a=0}^{d} p^a I_{m_{h,a}}, \qquad \operatorname{coker}(\lambda_d) \simeq \bigoplus_{a=1}^{d} (W(k)/p^a)^{\oplus m_{h,a}}. \tag{8.5}$$

These are module equivalences, not simultaneous diagonalizations of the algebra or congruences of quadratic forms. The multiplicity of a fixed exponent a is independent of d once $d \ge a$.

Proof. Write the trace Smith exponents as $a_1, \dots, a_{N_{h,d}}$. The Euler comparison gives $0 \le a_i \le d$. In Smith coordinates, $B_d/\mathcal{T}^d(B_d)$ has invariant exponents $d - a_i$, omitting zeros. By theorem 8.1 this quotient is $\mathcal{O}/I_d$, up to the fixed coefficient twist. Its invariant factors are given by theorem 7.2. For $a < d$, the exponent $d - a$ occurs $N_{h,a} - N_{h,a-1}$ times. The remaining trace exponent d occurs $N_{h,d} - N_{h,d-1}$ times, by subtracting these multiplicities from the rank $N_{h,d}$. □

Theorem 8.7 (All shallower levels in one endpoint algebra)**.** *For every $0 \le s \le d$,*

$$\tau_d(I_s) = \mathcal{T}^s(B_d), \qquad I_s = \tau_d^{-1}(\mathcal{T}^s(B_d)). \tag{8.6}$$

Thus membership in I_s can be tested by ordinary traces at any endpoint depth $d \ge s$:

$$x \in I_s \quad \Longleftrightarrow \quad \operatorname{Tr}^{\text{source}}_{B_d/W(k)}(\tau_d(x)b) \in p^s W(k) \text{ for every } b \in B_d.$$

Moreover, $\mathcal{T}^s(B_d) = (\bar{c}_d B_d : p^{d-s})$.

Proof. For $s = 0$ the assertion is immediate. For $1 \le s \le d$, expand the geometric recursion as $I_s = p^s\mathcal{O} + \sum_{t=0}^{s-1} p^{s-1-t} E_t$. The trace–Bockstein bound puts $\tau_d(E_t)$ in $\mathcal{T}^{t+1}(B_d)$ for $t < s \le d$. Hence $\tau_d(I_s) \subseteq \mathcal{T}^s(B_d)$. Since $E_d \subseteq I_d \subseteq I_s$, the first quotient has colength $\operatorname{length}(\mathcal{O}/I_s) = L_s$. By theorem 8.6, the second has colength

$$\sum_{a=0}^{s-1} (s-a)(N_{h,a} - N_{h,a-1}) = \sum_{a=0}^{s-1} N_{h,a} = L_s.$$

The inclusion is therefore equality, and taking inverse images proves (8.6). Finally, $p^{d-s}b \in \mathcal{T}^d(B_d)$ is equivalent to $\mathrm{Tr}(bB_d) \subseteq p^s W(k)$, by cancellation of p^{d-s} in $W(k)$. Use (8.1) to obtain the colon formula. □

Remark 8.8 (Explicit finite checks). At height two and $p = 2$, the endpoint polynomials in section B give trace matrices of ranks 3, 7, and 15. Their Smith valuation multisets are respectively $\{0, 1, 1\}$, $\{0, 1, 1, 2, 2, 2, 2\}$, and $\{0^1, 1^2, 2^4, 3^8\}$, where superscripts in the last multiset denote multiplicities. Their determinant valuations are 2, 10, and 34. These exact checks test the explicit endpoint models and the entire low-depth trace spectrum; they do not replace the all-height proof.

9. Height two: endpoint divisors and Hecke correspondences

The one-parameter geometry is governed by modular isogeny complexes and the Bruhat–Tits tree [32, 19, 37]. For the explicit calculation, work first over $k = \overline{\mathbf{F}}_p$ and fix Zhu's preferred height-two deformation data [44, Definition 2.24]. Put

$$W = W(k), \qquad \mathcal{O} = W[[h]], \qquad \mathfrak{m} = (p, h),$$

and fix the source section $h \mapsto 0$ in that coordinate. The same calculation applies after scalar extension of this coordinated model and this source. We do not identify this section with an arbitrary chosen lift. The intrinsic theorem for every perfect field and every source lift is obtained separately in section 9.9. The symbol h in this section is the deformation parameter; the height is fixed at two. Set

$$N_r = 1 + p + \cdots + p^r = \frac{p^{r+1} - 1}{p - 1}.$$

Theorem 9.1 (Rezk's height-two filtration package)**.** *The Witt filtration $\mathcal{O} = I_0 \supset I_1 \supset \cdots$ satisfies*

$$I_{r+1} \cap p\mathcal{O} = pI_r, \qquad I_1 = (p, h).$$

Moreover, I_{r+1}/pI_r is generated by a monic degree-N_r element reducing to h^{N_r}, and

$$\mathrm{length}_W(\mathcal{O}/I_r) = \sum_{t=0}^{r-1} N_t.$$

Proof. The saturation identity and the maximal-ideal formula are [33, Theorem 7.1]; the length is the associated-graded calculation in [33, Section 8], also recovered by theorems 7.1 and 7.2. At height two, proposition 4.3 gives a principal endpoint ideal with a monic Weierstrass generator of degree N_r and reduction h^{N_r}. The recursion in theorem 4.2 shows that its class generates I_{r+1}/pI_r. □

Theorem 9.2 (Zhu's all-prime height-two coordinate)**.** *There is a subgroup parameter α for which the universal order-p subgroup algebra is*

$$\mathcal{A} = \mathcal{O}[\alpha]/(w(h,\alpha)),$$

where

$$w(h,\alpha) = (\alpha - p)(\alpha + (-1)^p)^p - (h - p^2 + (-1)^p)\alpha.$$

It is finite free with basis $1, \alpha, \dots, \alpha^p$, and the total additive operation is

$$\psi(x) = \sum_{i=0}^{p} Q_i(x)\alpha^i.$$

The operation Q_0 represents the Frobenius class and

$$Q_0(h) \equiv h^p \pmod{p}. \tag{9.1}$$

If β is the dual subgroup parameter, then

$$\alpha\beta = (-1)^{p-1}p, \qquad B = f_p(\alpha) + \beta, \qquad X = f_p(\beta) + \alpha,$$

with

$$f_p(T) = p^2 - (-1)^p + \frac{(T-p)(T+(-1)^p)^p - (-1)^{p-1}p}{T}. \tag{9.2}$$

The additive operation ring is generated by the Q_i and satisfies Zhu's quadratic Adem relations.

Proof. The subgroup-ring presentation and the order-p operation are [44, Theorems 1.3 and 1.8] for the preferred data fixed above; the operation-ring presentation is [44, Theorem 1.9]. The dual parameter and its normalization are given in [44, Remark 1.6 and Equation (2.25)]. Solving $w(B,\alpha) = 0$ for B, using $\alpha\beta = (-1)^{p-1}p$, gives $B = f_p(\alpha) + \beta$ with (9.2). Applying the normalized duality exchanges α and β and gives the target formula $X = f_p(\beta) + \alpha$. □

Lemma 9.3 (Frobenius congruence for Q_0)**.** *For every $x \in \mathcal{O}$,*

$$Q_0(x) \equiv x^p \pmod{p}.$$

Proof. The operation Q_0 represents Rezk's Frobenius class. The displayed congruence is the weight-p congruence condition for the torsion-free $\mathbb{P}$-algebra $\mathcal{O}$ [30, Theorem A and Section 10]. Equation (9.1) is its specialization to the deformation parameter h. □

9.1. **Scheme-theoretic endpoint equations.** Let $\mathcal{O}_r$ be Strickland's finite-free ring classifying order-p^r subgroups of the universal deformation, with maps $s_r, t_r : \mathcal{O} \to \mathcal{O}_r$ [22, 39]. For the preferred zero source, put

$$\mathcal{B}_r = W\widehat{\otimes}_{\mathcal{O},s_r}\mathcal{O}_r, \qquad x_r = \tau_r(h) = t_r(h) \in \mathcal{B}_r.$$

This algebra is finite free of rank N_r over the source copy of W. Before using the preferred coordinates, we record a coefficient lemma that holds for arbitrary height-two source data.

Lemma 9.4 (Source characteristic polynomial and target equation)**.** *Let ℓ be any perfect field of characteristic p, let $R_\ell = W(\ell)[[h]]$, and fix any mixed-characteristic source section $\epsilon : R_\ell \to W(\ell)$. Form its endpoint algebra B_d^ϵ and put $x_d = \tau_d(h)$. Using the source coefficient structure, set*

$$P_d^\epsilon(T) = \det(T - m_{x_d} \mid B_d^\epsilon) \in W(\ell)[T].$$

If φ is Witt Frobenius, then the monic target-coefficient endpoint equation is

$$H_d^\epsilon(T) = \varphi^{-d}(P_d^\epsilon(T)), \qquad E_d^\epsilon = (H_d^\epsilon(h)). \tag{9.3}$$

Here Frobenius acts coefficientwise and fixes T. Both polynomials are monic of degree N_d and reduce to T^{N_d} modulo p.

Proof. The characteristic-p endpoint theorem gives $B_d^\epsilon/pB_d^\epsilon \simeq \ell[T]/(T^{N_d})$, with T the target parameter [33, Proposition 8.5]. Thus $1, x_d, \dots, x_d^{N_d-1}$ form a basis modulo p. Nakayama and equality of ranks make these powers an integral basis over the source copy of $W(\ell)$. Write $P_d^\epsilon(T) = \sum_i b_i T^i$. By remark 4.1,

$$\tau_d\left(\sum_i \varphi^{-d}(b_i)h^i\right) = \sum_i b_i x_d^i = 0.$$

The polynomial inside τ_d is monic and distinguished. Weierstrass division writes any element of R_ℓ as a multiple of H_d^ϵ plus a remainder $\sum_{i<N_d} a_i h^i$. If the remainder maps to zero, the power basis forces $\varphi^d(a_i) = 0$ for every i, hence $a_i = 0$. The special-fiber power basis also gives the stated reductions and degrees. □

Proposition 9.5 (Endpoint power basis)**.** *For the preferred zero source, the powers $1, x_r, \dots, x_r^{N_r-1}$ form a source W-basis of $\mathcal{B}_r$. There is a unique monic source characteristic polynomial P_r with*

$$\mathcal{B}_r \simeq W[X]/(P_r(X)), \qquad X \longmapsto x_r, \qquad P_r(X) \equiv X^{N_r} \pmod{p}.$$

The full scheme-theoretic target image has monic equation $H_r = \varphi^{-r}(P_r)$. In this preferred model, the integral construction in proposition 9.9 shows in addition that $P_r \in \mathbf{Z}_p[X]$, so $H_r = P_r$.

Proof. The power basis and the target-kernel assertion are the specialization of lemma 9.4. Multiplication by x_r has a cyclic vector 1, so its minimal and characteristic polynomials both have degree N_r and agree. The last assertion follows once the integral construction below is established; it is not used in that construction. □

For the universal order-p correspondence, retain the integral polynomial before completing in the source parameter.

Proposition 9.6 (Integral universal endpoint polynomial)**.** *Let $\delta_p = (-1)^{p-1}p$ and put*

$$w_p(B, \alpha) = \alpha(f_p(\alpha) - B) + \delta_p \in \mathbf{Z}_p[B, \alpha].$$

The algebra

$$\mathscr{A}_p^{\mathrm{univ}} = \mathbf{Z}_p[B,\alpha]/(w_p(B,\alpha))$$

is finite free of rank $p+1$ *over* $\mathbf{Z}_p[B]$*. In it define*

$$\beta = B - f_p(\alpha), \qquad X = f_p(\beta) + \alpha.$$

Then $\alpha\beta = \delta_p$*. The characteristic polynomial*

$$R_p(B,T) = \det(T - m_X \mid \mathscr{A}_p^{\mathrm{univ}}) \in \mathbf{Z}_p[B,T] \tag{9.4}$$

is the exact source–target endpoint equation. It is monic of degree $p+1$ *in each variable and satisfies* $R_p(B,T) = R_p(T,B)$*.*

Proof. The numerator divided by T in (9.2) has zero constant term, so $f_p \in \mathbf{Z}_p[T]$. Its leading term is T^p, and hence w_p is monic of degree $p+1$ in α; therefore $1, \alpha, \ldots, \alpha^p$ is a basis. Introducing $\beta = B - f_p(\alpha)$ identifies the algebra with

$$\mathbf{Z}_p[\alpha,\beta]/(\alpha\beta - \delta_p).$$

This is a domain. Swapping α and β interchanges B and X and realizes the normalized duality of [44, Remark 1.6].

After completion over $\mathbf{Z}_p[[B]]$, reduction modulo (p,B) gives $\alpha^{p+1} = 0$, $\beta = -\alpha^p$, and $X = \beta^p + \alpha = \alpha$. Here $p^2 \geq p+1$. Consequently $1, X, \ldots, X^p$ form a basis modulo (p,B); Nakayama makes them a basis of the completed algebra. They are therefore also independent over $\mathbf{Q}_p(B)$: a relation can be cleared of denominators and tested in that completed free algebra. Thus $R_p(B,T)$ is the minimal polynomial of X over $\mathbf{Q}_p(B)$, of degree $p+1$.

The kernel of $\mathbf{Q}_p[B,T] \to \mathscr{A}_p^{\mathrm{univ}}[1/p]$ is the principal prime ideal generated by this monic irreducible polynomial. The involution preserves this kernel, so $R_p(T,B) = cR_p(B,T)$ for a constant $c \in \mathbf{Q}_p^\times$. Applying the involution twice gives $c^2 = 1$. If $c = -1$, substitution $B = T$ gives zero; then $B - T$ divides the irreducible R_p, contradicting its degree $p+1 > 1$ in T. Thus $c = 1$. Symmetry gives degree $p+1$ and monicity in B as well. The identity holds integrally because the coefficients lie in $\mathbf{Z}_p$. The preferred source and target formulas of theorem 9.2 identify this relation with the geometric correspondence. □

After extension to W and completion in B, regard the universal order-p algebra as finite free over $W[[B]]$ through its source map.

Lemma 9.7 (Target power basis)**.** *The elements* $1, X, \ldots, X^p$ *form a* $W[[B]]$*-basis of the completed universal order-p algebra. Thus the base change of* $R_p(B,X) \in \mathbf{Z}_p[B,X]$ *is its unique monic target relation.*

Proof. Modulo (p,B) the source algebra is $k[\alpha]/(\alpha^{p+1})$ and $X = \alpha$, as in the proof of proposition 9.6. The target powers are therefore a basis after reduction. Nakayama and equality of ranks give the integral basis. Its unique monic degree-$(p+1)$ relation is the characteristic polynomial R_p. □

Lemma 9.8 (Normalized scalar backtracking)**.** *In the preferred correspondence, the dual of the quotient by an order-p subgroup D is the quotient by $F[p]/D$. Its composite with the original quotient is the preferred scalar isogeny $[\delta_p]$, with the specified marking isomorphism, and the final deformation parameter equals the initial one. Removing successive backtracking pairs therefore preserves the endpoint parameter.*

Proof. The subgroup and marking assertion is the normalized dual-isogeny identity of [44, Equation (2.16), Definition 2.24, and Equation (2.25)]. In the coordinate presentation, duality exchanges α and β, and hence exchanges

$$(B, X) = \big(f_p(\alpha) + \beta,\ f_p(\beta) + \alpha\big).$$

The dual step therefore returns the original parameter B exactly, not merely an isogenous formal group of the same degree. Repeated removal gives the assertion for any number of scalar backtracking pairs. The coefficient copy undergoes two Frobenius actions in such a pair; these preserve the reference zero section. Moreover the correspondence R_p has coefficients in $\mathbf{Z}_p$, so the same parameter relation is used at every edge. □

Proposition 9.9 (Exact-distance factors)**.** *For the preferred zero source there are unique monic exact-distance polynomials $C_r \in \mathbf{Z}_p[X]$, with $C_0 = X$, such that the source characteristic polynomials and target equations coincide and satisfy*

$$P_r(X) = H_r(X) = \prod_{j=0}^{\lfloor r/2 \rfloor} C_{r-2j}(X). \tag{9.5}$$

Their degrees are

$$d_0 = 1, \qquad d_r = (p+1)p^{r-1} \quad (r \geq 1). \tag{9.6}$$

Proof. Trivialize the generic Tate module as $T \simeq \mathbf{Z}_p^2$. An order-p^r subgroup corresponds to an over-lattice $L \supseteq T$ with $[L : T] = p^r$. In a suitable basis, elementary divisors give

$$L = p^{-a}\mathbf{Z}_p e_1 \oplus p^{-b}\mathbf{Z}_p e_2, \qquad a \geq b \geq 0, \quad a + b = r.$$

Its homothety class is at distance $a - b = r - 2b$ from the root. Its primitive representative $p^b L$ contains T with cyclic quotient of order p^{r-2b}. Conversely, a vertex at distance $r - 2b$ determines exactly one such L at total index p^r. Thus the subgroup labels are the disjoint union of spheres of radii $r, r-2, \ldots$. By lemma 9.8, scalar backtracking identifies the endpoint parameter with that of the primitive representative. The generic source characteristic polynomial is consequently the product of the corresponding sphere polynomials. The sphere sizes are one at radius zero, $p + 1$ at radius one, and p times the preceding size thereafter, proving (9.6).

The sphere polynomials are integral over $\mathbf{Z}_p$. For a monic polynomial $F(B) = \prod_v (B - b_v)$ define

$$\mathcal{N}(F)(X) = (-1)^{(p+1)\deg F} \operatorname{Res}_B(R_p(B, X), F(B)).$$

The root formula for the resultant gives $\mathcal{N}(F) = \prod_v R_p(b_v, X)$. Thus $\mathcal{N}$ preserves monic integral polynomials and counts all neighbors of the labeled vertices, with multiplicity. At radius one the root occurs $p+1$ times; each radius-two vertex occurs once. For radius $r \geq 2$, each vertex at radius $r-1$ occurs p times and each at radius $r+1$ occurs once. The generic sphere polynomials therefore satisfy

$$\mathcal{N}(C_0) = C_1, \qquad \mathcal{N}(C_1) = C_0^{p+1} C_2, \qquad \mathcal{N}(C_r) = C_{r-1}^{p} C_{r+1} \quad (r \geq 2).$$

Start with $C_0 = X$ and $C_1 = R_p(0, X) \in \mathbf{Z}_p[X]$. Inductively divide the integral polynomial $\mathcal{N}(C_r)$ by the indicated monic previous factor. Monic division takes place in $\mathbf{Z}_p[X]$. Its remainder vanishes after the injective extension to $\mathrm{Frac}(W)$ by the generic incidence identity, so it already vanishes over $\mathbf{Z}_p$. This constructs every C_r integrally. Their product is the generic characteristic polynomial P_r; since both sides are monic over the torsion-free coefficient ring W, the equality is integral.

In particular $P_r \in \mathbf{Z}_p[X]$. Frobenius fixes its coefficients, so lemma 9.4 gives $H_r = P_r$. Uniqueness follows successively from $C_0 = H_0$, $C_1 = H_1$, and the monic quotient $C_r = H_r / H_{r-2}$. This argument counts subgroup labels; it does not assume that unframed isomorphism classes separate them. At a fixed depth their separation also follows from the target power basis and generic étaleness of the endpoint algebra. □

Remark 9.10 (Polynomial descent, not descent of all choices). The construction proves that the displayed polynomial identities have coefficients in $\mathbf{Z}_p$ and remain valid after scalar extension of the preferred model. It does not assert that an arbitrary formal group, coordinate, source section, and system of isogeny markings descend together to this model. The intrinsic ideal equality will be transported by lemmas 9.33 and 9.34 instead.

9.2. **The endpoint correspondence modulo p^2.** The first step is an explicit but uniform first-order expansion of Zhu's coordinate.

Definition 9.11. *Let $A_p(T), L_p(T) \in \mathbf{F}_p[T]$ be defined by*

$$A_2(T) = 1, \qquad L_2(T) = 0,$$

and, for odd p,

$$A_p(T) = 1 + \sum_{j=1}^{p-2} j^{-1} T^{j+1} - 2T^p, \qquad L_p(T) = \frac{1 - A_p(T)}{T}. \tag{9.7}$$

Whenever pA_p or pL_p occurs modulo p^2, arbitrary integral lifts of the displayed polynomials are understood; the result is independent of the chosen lifts.

Lemma 9.12 (First correction to Zhu's coordinate)**.** *One has*

$$f_p(T) \equiv T^p - pL_p(T) \pmod{p^2}. \tag{9.8}$$

Equivalently,

$$A_p(T) = 1 - TL_p(T).$$

Moreover, for odd p, $A_p(T) - 1$ is divisible by T^2.

Proof. Expand equation (9.2) by the binomial theorem modulo p^2. For $1 \le j \le p-1$ one uses

$$\frac{1}{p}\binom{p}{j} \equiv (-1)^{j-1} j^{-1} \pmod{p}.$$

After division by T, the constant and linear terms cancel against $p^2 - (-1)^p$ and the dual-product correction. The remaining coefficient is exactly the polynomial in equation (9.7). The case $p = 2$ follows by direct expansion of equation (9.2) modulo four. □

Define the symmetric polynomial

$$\begin{aligned} E_p(B, X) = A_p(B) + A_p(X) - 1 + B^p L_p(X) + X^p L_p(B) \\ - B^{p-1} X^{p-1}. \end{aligned} \tag{9.9}$$

Theorem 9.13 (Uniform endpoint formula modulo p^2)**.** *The integral endpoint correspondence satisfies*

$$R_p(B, X) \equiv (B - X^p)(B^p - X) + pE_p(B, X) \pmod{p^2}. \tag{9.10}$$

In particular,

$$R_p(0, X) \equiv X^{p+1} + pA_p(X) \pmod{p^2}. \tag{9.11}$$

Proof. Work in the universal source–target ring modulo p^2, generated by α, β with $\alpha\beta = \delta_p$. By lemma 9.12,

$$B \equiv \alpha^p + \beta - pL_p(\alpha), \qquad X \equiv \beta^p + \alpha - pL_p(\beta) \pmod{p^2}.$$

Every mixed binomial term in $(\beta^p + \alpha)^p$ or $(\alpha^p + \beta)^p$ contains both α and β; the binomial coefficient supplies one factor p and the relation $\alpha\beta = \delta_p$ supplies another. Hence

$$B - X^p \equiv \beta - \beta^{p^2} - pL_p(\alpha),$$

$$B^p - X \equiv \alpha^{p^2} - \alpha + pL_p(\beta) \pmod{p^2}.$$

Their product is divisible by p. Reduce its quotient by p modulo p. The special fiber

$$k[\alpha, \beta]/(\alpha\beta)$$

is the reduced union of the two coordinate axes. On the component $\beta = 0$, the quotient is

$$\alpha^{p^2-1} - 1 - (\alpha^{p^2} - \alpha)L_p(\alpha). \tag{9.12}$$

On the same component, substitute $B = \alpha^p$ and $X = \alpha$ into equation (9.9). Using $A_p = 1 - TL_p$, one obtains the negative of equation (9.12). The calculation on the component $\alpha = 0$ is symmetric. Thus the right side of equation (9.10) vanishes in the universal target algebra modulo p^2.

It is monic of degree $p + 1$ in X, and its reduction is

$$(B - X^p)(B^p - X),$$

the degree-$(p+1)$ endpoint equation on the two components of the special fiber. By the target power basis in lemma 9.7, a monic degree-$(p+1)$ polynomial which annihilates the universal target coordinate is unique. It is therefore the reduction of the exact characteristic polynomial $R_p(B, X)$ from proposition 9.6. Setting $B = 0$ gives equation (9.11), since $E_p(0, X) = A_p(X)$. □

9.3. **The Hecke transform and exact-distance endpoints.**

Definition 9.14 (Signed monic endpoint transform)**.** *Let $F(B) \in \mathbf{Z}_p[B]$ be monic of degree D. The endpoint transform is the signed finite-free norm*

$$\mathcal{T}_p(F)(X) = (-1)^{(p+1)D} \operatorname{Norm}_{\mathbf{Z}_p[B,X]/(R_p)\,/\,\mathbf{Z}_p[X]} F(B). \tag{9.13}$$

Equivalently,

$$\mathcal{T}_p(F)(X) = (-1)^{(p+1)D} \operatorname{Res}_B(R_p(B, X), F(B)),$$

the unique monic associate of the source resultant. This is the algebraic Hecke transform attached to the height-two modular isogeny correspondence [32, 45].

Lemma 9.15 (Normalization and multiplicativity of the endpoint transform)**.** *The polynomial $\mathcal{T}_p(F)$ is monic of degree $(p+1)D$, and the transform is multiplicative on monic polynomials. In particular,*

$$\mathcal{T}_p(B) = R_p(0, X) \tag{9.14}$$

at every prime.

Proof. Since $R_p(B, X)$ is monic of degree $p+1$ in B, the raw norm is the source resultant. After a faithfully flat extension write $F(B) = \prod_{j=1}^{D}(B - \beta_j)$. Multiplicativity of the raw norm gives

$$\operatorname{Norm}(F(B)) = \prod_{j=1}^{D} \operatorname{Norm}(B - \beta_j) = (-1)^{(p+1)D} \prod_{j=1}^{D} R_p(\beta_j, X).$$

Each factor $R_p(\beta_j, X)$ is monic of degree $p+1$ in X. The sign in (9.13) therefore makes the transform monic of degree $(p+1)D$. Because its exponent is additive in D, this normalization preserves multiplicativity and descends from the splitting extension. For $F(B) = B$, the characteristic-polynomial identity gives

$$\operatorname{Norm}(B) = (-1)^{p+1} R_p(0, X),$$

and (9.14) follows. □

Remark 9.16. The sign in (9.13) is essential at $p = 2$. For the cubic correspondence

$$R_2(B, X) = B^3 - B^2X^2 + 9BX + X^3 - 54,$$

the raw norm of B is $-(X^3 - 54)$, whereas the signed monic transform is $\mathcal{T}_2(B) = X^3 - 54 = C_1(X)$.

Proposition 9.17 (Integral Hecke recursion)**.** *All exact-distance factors C_r and all-subgroup endpoint polynomials H_r are obtained by base change from polynomials in $\mathbf{Z}_p[X]$. They satisfy*

$$\mathcal{T}_p(C_0) = C_1, \qquad \mathcal{T}_p(C_1) = C_0^{p+1} C_2,$$

and, for $r \geq 2$,

$$\mathcal{T}_p(C_r) = C_{r-1}^{p} C_{r+1}.$$

In the first identity C_0 is evaluated in the source variable, so $C_0(B) = B$.

Proof. The signed resultant $\mathcal{N}$ used in the proof of proposition 9.9 is exactly the transform $\mathcal{T}_p$ of definition 9.14. For $C_0(B) = B$, lemma 9.15 gives $\mathcal{T}_p(C_0) = R_p(0, X) = C_1$. For the first sphere, the root is a neighbor of all $p + 1$ vertices, whereas each radius-two vertex occurs once. This gives $\mathcal{T}_p(C_1) = C_0^{p+1} C_2$. For radius $r \geq 2$, each vertex at radius $r - 1$ has p outward neighbors in the radius-r sphere, and each vertex at radius $r + 1$ has one inward neighbor. This gives the remaining identity. The monic-division argument of proposition 9.9 proves these identities over $\mathbf{Z}_p$, not only generically, and proves integrality of every C_r and H_r. The coefficient-twist issue disappears here precisely because these preferred-source polynomials lie in $\mathbf{Z}_p[X]$. □

Lemma 9.18 (First variation of the Hecke transform)**.** *Let $D \geq 1$ and suppose*

$$F(B) \equiv B^D + pG(B) \pmod{p^2}.$$

Then

$$\begin{aligned} \mathcal{T}_p(F)(X) \equiv X^{(p+1)D} + pX^D G(X^p) \\ + pD\, X^{(p+1)(D-1)} A_p(X) \pmod{p^2}. \end{aligned} \tag{9.15}$$

Proof. Introduce a square-zero parameter ϵ over $\mathbf{F}_p$. The special-fiber correspondence factors as

$$R_{p,0}(B, X) = (B - X^p)(B^p - X).$$

All branch products in this proof use the sign normalization of definition 9.14. On the rank-p branch the raw norm carries the factor $(-1)^{(p+1)D}$; the defining sign cancels it. The normalized product therefore has leading monomial $X^{(p+1)D}$, and the linear terms below are computed in that monic normalization.

For the variation of F, signed norm multiplicativity gives two factors. The factor associated with $B - X^p$ is

$$X^{pD} + \epsilon G(X^p).$$

The factor associated with $B^p - X$ has no first variation: in the purely inseparable rank-p algebra $B^p = X$, the norm of $B^D + \epsilon G(B)$ is its pth power, and $\epsilon^p = 0$. Multiplying by the unperturbed factor X^D gives the term $X^D G(X^p)$.

For the variation of the correspondence, set $F = B^D$. By lemma 9.15,

$$\mathcal{T}_p(B^D) = \mathcal{T}_p(B)^D = R_p(0, X)^D.$$

Using theorem 9.13,

$$R_p(0, X) = X^{p+1} + pA_p(X),$$

so the first variation is

$$DX^{(p+1)(D-1)}A_p(X).$$

The two first variations add; their product is of order p^2 and vanishes. □

The exact-distance polynomials C_r are the base changes of the integral polynomials constructed in proposition 9.17. Their recursion is

$$\mathcal{T}_p(C_0) = C_1, \qquad \mathcal{T}_p(C_1) = C_0^{p+1}C_2, \tag{9.16}$$

$$\mathcal{T}_p(C_r) = C_{r-1}^p C_{r+1} \qquad (r \geq 2). \tag{9.17}$$

No additional vertical factor occurs, because the quotient in the integral monic division was already identified from the generic incidence relation.

Define

$$\mathcal{B}_p(T) = T^{p^2-1}A_p(T) + A_p(T^p). \tag{9.18}$$

Theorem 9.19 (Uniform exact-distance congruence)**.** *One has*

$$C_1(X) \equiv X^{p+1} + pA_p(X) \pmod{p^2}, \tag{9.19}$$

and, for every $r \geq 2$,

$$C_r(X) \equiv X^{d_r} + p\mathcal{B}_p(X^{p^{r-2}}) \pmod{p^2}. \tag{9.20}$$

Proof. Equation (9.19) is exactly (9.11). Apply lemma 9.18 to C_1, whose degree is $p+1$. Since $p(p+1) \equiv p \pmod{p^2}$ and $\mathcal{T}_p(C_1) = X^{p+1}C_2$, division by X^{p+1} gives

$$C_2(X) \equiv X^{p(p+1)} + p\left(X^{p^2-1}A_p(X) + A_p(X^p)\right).$$

This is equation (9.20) at $r = 2$.

For $r \geq 2$, the degree d_r is divisible by p, so the final term in equation (9.15) vanishes modulo p^2. Moreover

$$C_{r-1}(X)^p \equiv X^{pd_{r-1}} = X^{d_r} \pmod{p^2}.$$

After dividing the recursion equation (9.17) by this factor, the correction polynomial is obtained from the preceding one by $X \mapsto X^p$. Induction gives equation (9.20). □

Corollary 9.20 (Eisenstein exact-distance endpoints)**.** *For every $r \geq 1$, the polynomial C_r is Eisenstein over W. It follows that C_r is irreducible over* $\mathrm{Frac}(W)$ *and $W[h]/(C_r)$ is a totally ramified extension of W of degree*

$$d_r = (p+1)p^{r-1}.$$

Proof. The reduction of C_r is h^{d_r}. Hence every nonleading coefficient is divisible by p. The formulas in equations (9.19) and (9.20) show that the constant term is p modulo p^2, because $A_p(0) = \mathcal{B}_p(0) = 1$. This is exactly Eisenstein's criterion. □

Corollary 9.21 (Frobenius self-similarity modulo p^2)**.** *For every $r \geq 2$,*

$$C_{r+1}(X) \equiv C_r(X^p) \pmod{p^2}.$$

Proof. Use $d_{r+1} = pd_r$ and compare the two instances of equation (9.20). □

9.4. **All-subgroup endpoints and the simultaneous carry.** The all-subgroup endpoint polynomial is the scheme-theoretic polynomial H_r of proposition 9.5, equivalently the product (9.5). Put

$$P_{r-1}(X) = \prod_{i=0}^{r-1} C_i(X), \qquad S_r = \deg P_{r-1} = \frac{(p+1)p^{r-1} - 2}{p-1}.$$

Proposition 9.22 (First correction of the all-subgroup endpoint)**.** *Modulo p^2 one has*

$$H_s(X) \equiv X^{N_s} + ph_s(X), \tag{9.21}$$

where

$$h_s(X) = \sum_{j=0}^{s-1} X^{N_s - d_{j+1}} A_p(X^{p^j}). \tag{9.22}$$

Proof. Expand equation (9.5) to first order in p. The correction of C_1 is $A_p(X)$. For $t \geq 2$, the correction of C_t is

$$X^{(p^2-1)p^{t-2}} A_p(X^{p^{t-2}}) + A_p(X^{p^{t-1}}).$$

After multiplication by the leading monomials of the other factors, the first summand is shifted by

$$N_s - d_t + (p^2-1)p^{t-2} = N_s - d_{t-1},$$

and the second by $N_s - d_t$. The adjacent pairs, together with the C_1 correction in odd parity, cover $j = 0, \ldots, s-1$ exactly once. □

Theorem 9.23 (Simultaneous endpoint carry)**.** *For every $r \geq 1$ and every $0 \leq i < r$,*

$$C_r(X) \equiv p \pmod{(p^2, C_i(X))}. \tag{9.23}$$

More precisely, if p is odd, there is a polynomial $Q_r(X)$ over $\mathbf{Z}_p/p^2$ such that

$$C_r(X) - p \equiv P_{r-1}(X)Q_r(X) \pmod{p^2}, \tag{9.24}$$

and $Q_r(X) \equiv X^{d_r - S_r} \pmod{p}$. At $p = 2$ one has

$$C_r(X) - 2 \equiv X^2 P_{r-1}(X) \pmod{4}. \tag{9.25}$$

In particular the weaker all-subgroup form

$$C_r(X) - p \in (p^2, H_{r-1}(X)) \tag{9.26}$$

holds at every prime.

Proof. For odd p, write

$$C_i = X^{d_i} + pg_i \pmod{p^2}, \qquad g_1 = A_p(X), \quad g_i = \mathcal{B}_p(X^{p^{i-2}}) \quad (i \geq 2).$$

We have

$$S_r = \frac{(p+1)p^{r-1} - 2}{p-1}, \qquad d_r - S_r \geq 0.$$

The case $r = 1$ is immediate: $P_0 = X$ and $A_p(X) - 1$ is divisible by X, so $Q_1 = X^p + p(A_p(X) - 1)/X$ gives (9.24).

Assume $r \geq 2$. The first-order expansion of $P_{r-1}(X)(X^{d_r - S_r} + pq(X))$ is

$$X^{d_r} + p\left(\sum_{i=1}^{r-1} g_i(X) X^{d_r - d_i} + X^{S_r} q(X)\right).$$

Thus it suffices to show that

$$D_r(X) = g_r(X) - 1 - \sum_{i=1}^{r-1} g_i(X) X^{d_r - d_i} \tag{9.27}$$

is divisible by X^{S_r} over $\mathbf{F}_p$. Since $A_p(T) - 1$ starts in degree at least two and $p^2 - 1 \geq 2p$ for odd p, every nonconstant term of $g_r - 1$ has degree at least $2p^{r-1}$. Every term in the sum has degree at least $d_r - d_{r-1} = (p^2 - 1)p^{r-2}$. Both bounds are at least S_r:

$$(p-1)(2p^{r-1} - S_r) = (p-3)p^{r-1} + 2 \geq 0,$$
$$(p-1)((p^2-1)p^{r-2} - S_r) = (p+1)(p^2 - 3p + 1)p^{r-2} + 2 \geq 0.$$

Hence $D_r = X^{S_r} q$. Taking an integral lift of q and setting $Q_r = X^{d_r - S_r} + pq$ proves (9.24).

For $p = 2$, put $d_i = 3 \cdot 2^{i-1}$ for $i \geq 1$. The input congruences are

$$C_1 \equiv X^3 + 2, \qquad C_i \equiv X^{d_i} + 2(X^{d_{i-1}} + 1) \pmod{4} \quad (i \geq 2).$$

The base case is separate:

$$C_1 - 2 \equiv X^3 = X^2 P_0 \pmod{4}.$$

For $r \geq 2$, $2 + \sum_{i=0}^{r-1} d_i = d_r$, with $d_0 = 1$. At most one correction contributes modulo four, so

$$X^2 P_{r-1} \equiv X^{d_r} + 2\left[X^{d_r - d_1} + \sum_{i=2}^{r-1} (X^{d_r - d_i + d_{i-1}} + X^{d_r - d_i})\right] \pmod{4}.$$

Inside the brackets work over $\mathbf{F}_2$. Since $d_i = 2d_{i-1}$, the first term at index i equals $X^{d_r - d_{i-1}}$ and cancels the second term at the preceding index. The initial $X^{d_r - d_1}$ starts the cancellation, leaving only $X^{d_r - d_{r-1}} = X^{d_{r-1}}$. The expression is therefore $X^{d_r} + 2X^{d_{r-1}} \equiv C_r - 2 \pmod{4}$. For $r = 2$ the sum is empty and the same conclusion holds.

Every C_i with $i < r$ divides P_{r-1}, so both forms give (9.23). Since H_{r-1} also divides P_{r-1}, they give (9.26). □

Corollary 9.24 (Functorial form of the carry)**.** *Let R be any $\mathbf{Z}_p$-algebra and let $z \in R$. Evaluation of the integral polynomial identity gives*

$$C_r(z) - p \in p^2R + H_{r-1}(z)R.$$

For odd p the stronger containment with $P_{r-1}(z)$ holds, while for $p = 2$ one has

$$C_r(z) - 2 \equiv z^2 P_{r-1}(z) \pmod{4R}.$$

Thus the carry is an identity of the universal endpoint polynomials, not only a congruence after substituting the deformation parameter h.

9.5. **Pairwise exact-distance intersections.**

Theorem 9.25 (Uniform pairwise intersection law)**.** *For every $0 \le i < j$,*

$$(C_i, C_j) = (C_i, p) \subseteq \mathcal{O}. \tag{9.28}$$

It follows that

$$\mathcal{O}/(C_i, C_j) \simeq k[[h]]/(h^{d_i}), \tag{9.29}$$

so the two Cartier divisors meet properly and Tor-independently, with intersection multiplicity d_i. Moreover

$$\mathrm{Res}(C_i, C_j) = p^{d_i} u_{ij}, \qquad u_{ij} \in W^\times, \qquad u_{ij} \equiv 1 \pmod{p}. \tag{9.30}$$

Proof. By theorem 9.23, in $\mathcal{O}/(C_i)$ one has

$$C_j = p(1 + pa)$$

for some a. The factor $1 + pa$ is a unit, which proves equation (9.28). Since $C_i \equiv h^{d_i} \pmod{p}$, the quotient is equation (9.29). The image of C_j is a non-zero-divisor because $\mathcal{O}/(C_i)$ is finite flat over W, so the intersection is Tor-independent. Taking the norm of $C_j = p(1 + pa)$ in the rank-d_i finite free W-algebra $W[h]/(C_i)$ proves equation (9.30). □

9.6. **The geometric filtration.** Define

$$J_0 = \mathcal{O}, \qquad J_{r+1} = pJ_r + (H_r(h)). \tag{9.31}$$

Theorem 9.26 (Intersection recursion and staircase generators)**.** *For every $r \ge 1$,*

$$J_r = (H_{r-1}, H_r). \tag{9.32}$$

For $r \ge 2$ one has the sharper elementary recursion

$$(H_{r-1}, H_r) = (H_{r-1}, pH_{r-2}). \tag{9.33}$$

Equivalently,

$$J_r = (p^r, p^{r-1}H_0, p^{r-2}H_1, \ldots, pH_{r-2}, H_{r-1}). \tag{9.34}$$

Proof. For $r = 1$, the carry for $C_1 = H_1$ gives $(H_0, H_1) = (h, p) = J_1$. For $r \ge 2$, write

$$C_r = p + H_{r-1}Q_r + p^2 A_r$$

using the weaker form equation (9.26). Since $H_r = C_r H_{r-2}$, reduction modulo H_{r-1} gives

$$H_r = pH_{r-2}(1 + pA_r).$$

The parenthetical factor is a unit, proving equation (9.33). Induction now gives

$$(H_{r-1}, H_r) = (H_{r-1}) + p(H_{r-2}, H_{r-1}) = (H_{r-1}) + pJ_{r-1} = J_r.$$

Expanding the recursion yields equation (9.34). □

Corollary 9.27 (p-saturation)**.** *For every $r \geq 1$,*

$$J_r \cap p\mathcal{O} = pJ_{r-1}. \tag{9.35}$$

Equivalently, $(J_r : p) = J_{r-1}$.

Proof. The inclusion from right to left is immediate. Write $z = pu + H_{r-1}v \in J_r \cap p\mathcal{O}$ with $u \in J_{r-1}$. Reducing modulo p gives $h^{N_{r-1}}\bar{v} = 0$ in the domain $k[[h]]$, hence $v = pv'$. Since $H_{r-1} \in J_{r-1}$ by equation (9.32) applied at $r-1$ (and trivially for $r = 1$, when $J_0 = \mathcal{O}$), one has $z = p(u + H_{r-1}v') \in pJ_{r-1}$. The colon statement follows from p-torsion-freeness of $\mathcal{O}$. □

Perform monic Euclidean division

$$H_r = B_r H_{r-1} + pR_r, \qquad \deg R_r < N_{r-1}. \tag{9.36}$$

The remainder is integral because both endpoint polynomials reduce to pure powers of h.

Theorem 9.28 (All-prime Euclidean carry)**.** *For every $r \geq 1$,*

$$R_r \in J_{r-1}.$$

For $r \geq 2$,

$$R_r \equiv H_{r-2} \pmod{pJ_{r-2}}. \tag{9.37}$$

Proof. Both H_r and $B_r H_{r-1}$ lie in J_r, so $pR_r \in J_r \cap p\mathcal{O} = pJ_{r-1}$. Since $\mathcal{O}$ is p-torsion-free, $R_r \in J_{r-1}$. The carry gives $R_r \equiv H_{r-2} \pmod{p}$. Write $R_r = pu + H_{r-2}v$ with $u \in J_{r-2}$. Reduction modulo p gives $h^{N_{r-2}}(\bar{v} - 1) = 0$, hence $v = 1 + pw$, which proves equation (9.37). □

9.7. **Invariant factors of the height-two quotients.** We give a direct calculation for J_r from corollary 9.27. Once $J_r = I_r$ is identified below, this recovers the height-two case of theorems 7.1 and 7.2.

Put $A_r = \mathcal{O}/J_r$.

Theorem 9.29 (Layer decomposition and Smith form)**.** *For $0 \leq s < r$, multiplication by p^s induces a canonical identification*

$$p^s A_r / p^{s+1} A_r \simeq k[[h]]/(h^{N_{r-s-1}}). \tag{9.38}$$

It follows that the complete finite W-module structure is

$$A_r \simeq \bigoplus_{e=1}^{r} (W/p^e)^{\oplus p^{r-e}}. \tag{9.39}$$

In particular

$$\operatorname{length}_W(A_r) = \sum_{e=1}^{r} ep^{r-e} = \frac{p(p^r-1) - r(p-1)}{(p-1)^2}. \tag{9.40}$$

Proof. Repeated use of $(J_m : p) = J_{m-1}$ gives $(J_r : p^s) = J_{r-s}$. Hence multiplication by p^s identifies $\mathcal{O}/J_{r-s}$ with $p^s A_r$. Under this identification the next submodule corresponds to multiplication by p, and therefore

$$p^s A_r / p^{s+1} A_r \simeq A_{r-s}/pA_{r-s} \simeq k[[h]]/(h^{N_{r-s-1}}).$$

For a finite module over the DVR W, the dimension of $p^s A_r/p^{s+1}A_r$ is the number of invariant factors of exponent strictly larger than s. It equals N_{r-s-1}. Thus the multiplicity of W/p^e is

$$N_{r-e} - N_{r-e-1} = p^{r-e},$$

with $N_{-1} = 0$. Summing the lengths gives equation (9.40). □

Corollary 9.30 (Complete intersection, Gorenstein property, and socle)**.** *For every $r \geq 1$, A_r is an Artinian complete intersection and hence Gorenstein. Its socle is one-dimensional over k and is generated by the class of p^{r-1}.*

Proof. The regular local ring $\mathcal{O}$ has dimension two and $J_r = (H_{r-1}, H_r)$ has finite colength, so its two generators form a regular sequence. The staircase presentation shows that $p^r = 0$ and $p^{r-1}h = 0$ in A_r, while equation (9.39) shows that $p^{r-1} \neq 0$. The Gorenstein socle has k-dimension one, so this class generates it. □

Corollary 9.31 (Resultant intersection formula)**.** *For every $r \geq 1$,*

$$v_p \operatorname{Res}(H_{r-1}, H_r) = \operatorname{length}_W(A_r) = \frac{p(p^r-1) - r(p-1)}{(p-1)^2}. \tag{9.41}$$

Proof. The colength of the monic complete intersection (H_{r-1}, H_r) is the valuation of its resultant. Apply theorems 9.26 and 9.29. □

9.8. **Identification for the preferred source.** For the preferred source, proposition 9.9 gives $E_r = (H_r)$ with $H_r \in \mathbf{Z}_p[X]$. Hence theorem 4.2 gives

$$I_{r+1} = pI_r + (H_r).$$

Starting with $I_0 = J_0 = \mathcal{O}$, it identifies the Witt filtration with the geometric filtration of section 9.6.

Proposition 9.32 (Preferred-source intersection)**.** *In the preferred coordinate with source $h = 0$, and after scalar extension of that coordinated model,*

$$I_r = (H_{r-1}, H_r) \qquad (r \geq 1).$$

Proof. The recursion just displayed identifies I_r with J_r. Apply theorem 9.26. The endpoint polynomial identities commute with scalar extension, and the all-height recursion identifies the resulting filtration with the Witt filtration over the extended field. □

9.9. **Change of source and faithfully flat descent.** We remove the preferred-source restriction. Let k again be any perfect field of characteristic p and $R = W(k)[[h]]$. The all-height recursion has already been proved; the following argument is a consequence of it, not an input to its proof.

Lemma 9.33 (Transfer between source families)**.** *Suppose two endpoint families have generators e_d and f_d satisfying*

$$I_{d+1} = pI_d + (e_d) = pI_d + (f_d), \qquad (\overline{e}_d) = (\overline{f}_d) = (h^{N_d}) \subset k[[h]].$$

If $I_r = (e_{r-1}, e_r)$ for one family, then $I_r = (f_{r-1}, f_r)$ for the other.

Proof. Since R/I_r has finite length, I_r has height two. Its displayed two generators are minimal: otherwise it would be principal, contradicting the principal ideal theorem. Their classes form a basis of $I_r/\mathfrak{m}_R I_r$. The recursion at depth $r+1$ permits a decomposition

$$f_r = ve_r + pz, \qquad z \in I_r.$$

Modulo p, both e_r and f_r generate (h^{N_r}); cancellation in $k[[h]]$ shows that $\overline{v}$ is a unit, hence $v \in R^\times$. Since $pz \in \mathfrak{m}_R I_r$, the class of f_r is a nonzero multiple of the class of e_r.

Also $f_{r-1} \in I_r$, so write

$$f_{r-1} = ae_{r-1} + be_r.$$

Reducing modulo p and cancelling $h^{N_{r-1}}$ shows that $\overline{a}$ has nonzero constant term: the other summand has the strictly positive extra order $N_r - N_{r-1}$. Thus a is a unit. Relative to the basis $[e_{r-1}], [e_r]$, the classes $[f_{r-1}], [f_r]$ have a triangular coefficient matrix with nonzero diagonal. They span $I_r/\mathfrak{m}_R I_r$. Nakayama applied to $I_r/(f_{r-1}, f_r)$ proves the conclusion. □

Lemma 9.34 (Perfect-field base change)**.** *Let $k \subseteq k'$ be an extension of perfect fields. Transport the height-two formal group and a chosen source section ϵ to k'. In compatible parameters, put*

$$R = W(k)[[h]], \qquad R' = W(k')[[h]].$$

Then $R \to R'$ is faithfully flat, and

$$E_d^\epsilon R' = E_d^{\epsilon'}, \qquad I_r R' = I'_r. \tag{9.42}$$

An isomorphism of the special-fiber formal groups transports these ideals through the induced isomorphism of their deformation rings.

Proof. Both rings are Noetherian regular local rings of dimension two. Moreover $\mathfrak{m}_R R' = \mathfrak{m}_{R'}$ and the closed fiber is k'. Miracle flatness applies since the target is Cohen–Macaulay and

$$\dim R' = \dim R + \dim(R'/\mathfrak{m}_R R') = 2 + 0;$$

see [38, Tag 00R4]. The map is flat, and a flat local map is faithfully flat.

The finite subgroup moduli and quotient targets commute with this extension. In the source coefficient structure the endpoint algebra becomes $B_d^\epsilon \widehat{\otimes}_{W(k)} W(k')$. Its source characteristic polynomial is obtained by extending

the coefficients of P_d^ϵ. Witt Frobenius commutes with the field extension. Hence lemma 9.4 identifies its target equation with the scalar extension of H_d^ϵ, proving the endpoint-ideal equality in (9.42).

For the Witt ideals no additional coordinate-naturality assertion is needed. Apply theorem 4.2 before and after extension. Starting with $I_0R' = R' = I'_0$, induction gives

$$I_{r+1}R' = p(I_rR') + E_r^{\epsilon'} = pI'_r + E_r^{\epsilon'} = I'_{r+1}.$$

An isomorphism of the special-fiber formal groups induces an isomorphism of deformation functors and subgroup-quotient functors. It therefore transports the endpoint ideals and their source sections. Applying the same recursion then transports the Witt ideals as well. □

Theorem 9.35 (Height-two geometric Witt theorem at all primes)**.** *Let Γ/k be a one-dimensional formal group of height two over a perfect field, and choose any mixed-characteristic source lift over $W(k)$. Write $E_r = E_r^\epsilon$, and let $H_r = H_r^\epsilon$ be its monic target-coefficient endpoint equation from lemma 9.4. For every $r \geq 0$,*

$$I_{r+1} = pI_r + E_r = pI_r + (H_r).$$

For every $r \geq 1$,

$$I_r = E_{r-1} + E_r = (H_{r-1}, H_r).$$

Equivalently, if $Z_r = V(E_r) \subset \operatorname{Spf} R$, then $V(I_r) = Z_{r-1} \cap Z_r$ scheme-theoretically.

Proof. The recursion is theorem 4.2. Extend the coefficient field to an algebraic closure k'. The classification of one-dimensional formal groups over an algebraically closed field by their finite height identifies $\Gamma_{k'}$ with the scalar extension of a preferred height-two model over $\overline{\mathbf{F}}_p$; see [14] and [44, Remark 2.26]. Transport the deformation ring along this isomorphism.

There are now two source sections in the same ring $W(k')[[h]]$: the transported original source and the preferred zero source. They are not assumed equal. For the reference source, proposition 9.32 proves the consecutive-endpoint identity. Both endpoint families satisfy theorem 4.2, and their reductions generate (h^{N_d}) by lemma 9.4. After a parameter change this statement holds up to units, which is sufficient for lemma 9.33. That lemma transfers the ideal identity to the transported original source.

Returning to the original coordinates after extension, we have

$$(E_{r-1} + E_r)R' = I_rR'$$

by lemma 9.34. The inclusion $E_{r-1} + E_r \subseteq I_r$ follows from the all-height recursion and endpoint carry. Its quotient in I_r becomes zero after the faithfully flat extension $R \to R'$, hence was already zero. Taking formal closed subschemes gives the final assertion. □

Remark 9.36 (Scope of the explicit factors)*.* The preceding theorem concerns endpoint ideals for any chosen source. It does not say that the polynomials

themselves are source independent, nor that an arbitrary-source H_r has a same-source factorization $C_rC_{r-2}\cdots$. The explicit f_p, R_p, C_r and their pairwise carry and intersection formulas above remain attached to the preferred coordinate and zero source. What transfers is the intrinsic equality $I_r = E_{r-1} + E_r$.

Corollary 9.37 (Exact mixed-characteristic thickness)**.** *For the source data of theorem 9.35 and every $r \geq 1$,*

$$R/I_r \simeq \bigoplus_{e=1}^{r} (W(k)/p^e)^{\oplus p^{r-e}}$$

as $W(k)$-modules. Moreover,

$$\operatorname{length}_{W(k)}(R/I_r) = v_p \operatorname{Res}(H_{r-1}, H_r) = \frac{p(p^r-1) - r(p-1)}{(p-1)^2}.$$

Each quotient is an Artinian Gorenstein complete intersection, with socle generated by p^{r-1}.

Proof. The module decomposition and length follow from theorem 7.2 at height two. The monic distinguished polynomials H_{r-1}, H_r generate the finite-colength ideal I_r by theorem 9.35. Multiplication by H_r in the finite free algebra $W(k)[h]/(H_{r-1})$ has cokernel R/I_r and determinant $\operatorname{Res}(H_{r-1}, H_r)$. The two generators form a regular sequence in the two-dimensional regular local ring R, so the quotient is Gorenstein. Repeated use of $pI_t \subseteq I_{t+1}$ gives $p^{r-1}\mathfrak{m}_R \subseteq I_r$, because $I_1 = \mathfrak{m}_R$. The quotient ring has exponent exactly p^r by its module decomposition, so the class of p^{r-1} is nonzero. It lies in the socle, which has dimension one, and therefore generates it. □

10. Ramification and arithmetic applications

Let K be a nonarchimedean local field with uniformizer π, residue cardinality q, and let F_{LT} be a Lubin–Tate formal $\mathcal{O}_K$-module for π. In a Lubin–Tate coordinate write [23, 35]

$$f(T) = [\pi]_{F_{\mathrm{LT}}}(T) = \pi T + a_2 T^2 + \cdots + a_{q-1}T^{q-1} + T^q, \qquad a_i \in \pi\mathcal{O}_K.$$

Put $F_n = f^{\circ n}$.

Proposition 10.1 (Universal discriminant divisibility)**.** *There is a polynomial $U_n(Y) \in \mathcal{O}_K[Y]$ such that*

$$\operatorname{Disc}_T(F_n(T) - Y) = \pi^{nq^n} U_n(Y), \qquad U_n(0) \in \mathcal{O}_K^{\times}.$$

Proof. Work in the finite free algebra

$$\mathscr{A}_n = \mathcal{O}_K[[Y]][T]/(F_n(T) - Y),$$

which has rank q^n over $\mathcal{O}_K[[Y]]$. Every coefficient of $f'(T)$ is divisible by π, so write

$$f'(T) = \pi u(T), \qquad u(T) \in \mathcal{O}_K[T], \qquad u(0) = 1.$$

The chain rule gives the identity in $\mathscr{A}_n$

$$F_n'(T) = \pi^n \prod_{j=0}^{n-1} u(F_j(T)).$$

Modulo (π, Y), each factor $u(F_j(T))$ has constant term one in the local Artin ring $k[T]/(T^{q^n})$, hence is a unit in $\mathscr{A}_n$. Taking the finite-free norm and using corollary 2.3 yields

$$\mathrm{Disc}_T(F_n(T) - Y) = \pm\pi^{nq^n} \prod_{j=0}^{n-1} \mathrm{Norm}_{\mathscr{A}_n/\mathcal{O}_K[[Y]]}(u(F_j(T))).$$

The product of norms is a unit in $\mathcal{O}_K[[Y]]$. Since the discriminant is a polynomial in Y, division by π^{nq^n} gives a polynomial $U_n(Y)$ whose constant term is a unit. □

Choose $a \in K$ with $v_K(a) = 1$ and a compatible branch $f(x_j) = x_{j-1}$, $x_0 = a$.

Theorem 10.2 (Exact branch ramification)**.** *For every $n \geq 1$,*

$$[K(x_n) : K] = q^n,$$
$$v_{K(x_n)}(\mathfrak{D}_{K(x_n)/K}) = nq^n, \qquad v_K \operatorname{Disc}(K(x_n)/K) = nq^n.$$

Proof. Modulo π, the series f is T^q. Once x_{j-1} is a uniformizer of $K(x_{j-1})$, the polynomial $f(T) - x_{j-1}$ is Eisenstein over that field. It is the minimal polynomial of x_j, so x_j is a uniformizer and $[K(x_j) : K(x_{j-1})] = q$. Induction gives the degree formula.

The relative different is generated by $f'(x_j)$. The constant term π has strictly smaller $K(x_j)$-valuation than every other term of $f'(x_j)$, so

$$v_{K(x_j)}(f'(x_j)) = q^j.$$

If $d_j = v_{K(x_j)}(\mathfrak{D}_{K(x_j)/K})$, transitivity of the different gives

$$d_j = q\, d_{j-1} + q^j, \qquad d_0 = 0.$$

Thus $d_j = jq^j$. The tower is totally ramified, so the discriminant has the same exponent over K. □

Corollary 10.3 (Root-discriminant contribution)**.** *With the notation of theorem 10.2,*

$$|\operatorname{Disc}(K(x_n)/K)|^{-1/[K(x_n):K]} = |\pi|^{-n}.$$

Proof. Divide the discriminant exponent nq^n by the degree q^n. □

Theorem 10.4 (Lubin–Tate affine Kummer group)**.** *Let*

$$K_n = K(F_{\mathrm{LT}}[\pi^n]), \qquad L_n = K_n(x_n), \qquad [\pi^n]_{F_{\mathrm{LT}}}(x_n) = a.$$

Then L_n is the splitting field of $F_n(T) - a$. Lubin–Tate reciprocity identifies

$$\mathrm{Gal}(K_n/K) \simeq (\mathcal{O}_K/\pi^n)^\times$$

through its action on $F_{\mathrm{LT}}[\pi^n]$. *If* $q > 2$, *then*

$$\mathrm{Gal}(L_n/K) \simeq (\mathcal{O}_K/\pi^n) \rtimes (\mathcal{O}_K/\pi^n)^\times.$$

Proof. The roots of $F_n(T) - a$ are $x_n +_{F_{\mathrm{LT}}} t$ with $t \in F_{\mathrm{LT}}[\pi^n]$, so adjoining x_n to the torsion field splits the polynomial. Lubin–Tate reciprocity gives the full unit action on the torsion module; see [35, Chapter V]. Thus the Galois image is a subgroup of the affine group $M_n \rtimes M_n^\times$, where $M_n = \mathcal{O}_K/\pi^n$, and it projects onto $M_n^\times$. Eisenstein irreducibility makes its action on the root torsor transitive.

Let $I \subseteq M_n$ be the translation subgroup. It is stable under all units. If $r \in M_n$ is a nonunit, then both 1 and $r-1$ are units, so for $x \in I$ one has $rx = x + (r-1)x \in I$. Thus I is an M_n-submodule, hence an ideal. If I were proper, the induced affine action on the nonzero quotient $\overline{M} = M_n/I$ would have trivial translation kernel and would be the graph of a cocycle $c : M_n^\times \to \overline{M}$. Choose a Teichmuller unit $\zeta \neq 1$; when $q > 2$, $\zeta - 1$ is a unit. Commutativity gives

$$(u-1)c(\zeta) = (\zeta - 1)c(u),$$

so $c(u) = (u-1)b$ for a fixed b. Translating the affine coordinate by b makes the unit group fix the origin. The induced action on $\overline{M}$ is then not transitive unless $\overline{M} = 0$, contradicting the transitivity inherited from the root torsor. Hence $I = M_n$. □

Corollary 10.5 (Local-to-global translation criterion)**.** *Let* K_0 *be a number field and let* (G, α, H) *be as in theorem 2.12. Let* $P \in G(K_0)$ *have trivial* H*-stabilizer. At a finite place* v, *put* $K = K_{0,v}$ *and suppose its residue cardinality is* $q > 2$. *Assume that the* entire *global* α*-adic Tate module and its compatible division torsors, restricted to a decomposition group at* v, *are equivariantly identified at every level with the rank-one Lubin–Tate module and division torsors of theorem 10.4. Assume that the local coordinate corresponding to* P *has valuation one. Then the global Kummer translation image is all of* $T_\alpha(G)$.

If the identification is available only after projection onto a local factor, the conclusion is only that the corresponding local affine projection is full. Containment in the global pure translation subgroup requires, additionally, lifts acting trivially on the linear part of the entire global module.

Proof. At each level the local theorem gives every translation in the Lubin–Tate module. Under the full-module and full-torsor identification, these are elements of the global Galois image with globally trivial linear part. To pass to the inverse limit, fix a compatible translation. Its sets of lifts with identity linear part at level n are nonempty nested closed subsets of the compact global Galois image. Their intersection gives the desired pure translation on the entire Tate module. The trivial stabilizer identifies quotient specialization with the Kummer torsor by theorem 2.12; no geometric connectedness of a finite separable fiber is required. For a factor projection, the same local

theorem gives surjectivity only after projection, since the complementary linear action of a lift need not vanish. □

The Chebotarev statements for reductions of such global families must be read with the constant-field convention of corollary 2.14: one fixes the Frobenius coset, or equivalently the degree class modulo the constant-field degree, before taking the limit.

11. Further directions

The recursion and Euler presentation are presentation-free. Together they determine the Witt filtration and force Koszul minimal resolutions for both endpoint and Witt ideals in every height.

Remark 11.1 (Uniform structure in higher height). By theorem 8.3, E_d has exactly $h-1$ minimal generators, and its quotient has Betti numbers $\binom{h-1}{i}$ over $\mathcal{O}$. Choosing such a regular sequence and appending an Euler lift gives the h generators of I_d in corollary 8.4. The general possibility of non-complete-intersection Gorenstein ideals [5] therefore does not occur for these fixed-source endpoint ideals. What remains unspecified is a canonical choice of the equations, or an intersection formula involving a bounded number of consecutive endpoint images in higher height.

The complete-intersection theorem leaves an explicit presentation problem: construct natural regular-sequence generators for E_d in height $h > 2$, preferably from subgroup flags. Any such construction must recover the Koszul ranks of theorem 8.3 and the mixed-characteristic layers of section 7.

A second problem is to identify the higher-height replacement for the height-two intersection formula. Candidate structures include derived intersections, Fitting ideals of flag algebras, and complexes built from subgroup flags. It would also be useful to compare the additive trace lattice with the higher multiplicative carries of the Lubin–Tate–Witt functor, and to quantify the information lost when ordered branch data are passed to symmetric norm and trace invariants.

Appendix A. The subgroup-width generating function

We record the subgroup-counting identity used throughout the paper. Let

$$N_{h,d} = \#\{D \leq (\mathbb{Q}_p/\mathbb{Z}_p)^h : |D| = p^d\}.$$

Put $V_j = (\mathbf{Q}_p/\mathbf{Z}_p)^j$. For a subgroup $D \subset V_h$ of order p^d, let its image in the last factor have order p^a, and put $E = D \cap V_{h-1}$. The last-factor image is the unique subgroup of order p^a, while E has order p^{d-a}. After fixing E and passing to the quotient, D/E has zero intersection with V_{h-1}/E and projects isomorphically to that image. It is therefore the graph of a homomorphism

$$\mathbf{Z}/p^a \longrightarrow V_{h-1}/E.$$

Conversely, the inverse image of each such graph determines one D. The group V_{h-1}/E is again divisible and p-primary of rank $h-1$; its p^a-torsion has $p^{a(h-1)}$ elements. Hence there are exactly that many homomorphisms, including one when $a = 0$. It follows that

$$N_{h,d} = \sum_{a=0}^{d} p^{a(h-1)} N_{h-1,d-a}, \qquad N_{1,d} = 1.$$

Multiplying generating series and inducting on h gives

$$\sum_{d\geq 0} N_{h,d} T^d = \prod_{j=0}^{h-1} (1 - p^j T)^{-1}. \tag{A.1}$$

Its coefficients are the Gaussian binomial coefficients [3, 25]:

$$N_{h,d} = \begin{bmatrix} d+h-1 \\ h-1 \end{bmatrix}_p.$$

Equivalently, these coefficients satisfy the q-Pascal recurrence $N_{h,d} = N_{h-1,d} + p^{h-1} N_{h,d-1}$ with the stated boundary values.

The numbers

$$L_r = \sum_{d=0}^{r-1} N_{h,d}$$

are the lengths of the Witt truncations. Their generating series is

$$\sum_{r\geq 1} L_r T^r = \frac{T}{(1-T)^2 \prod_{j=1}^{h-1} (1 - p^j T)}.$$

The extra factor $(1-T)^{-1}$ records the passage from a graded layer to the cumulative filtration.

Appendix B. Low-depth endpoint equations at height two

At height two and $p = 2$, write the deformation parameter as a. Use the preferred coordinate and its zero source from section 9. The one-step endpoint correspondence is

$$R(B, X) = B^3 - B^2 X^2 + 9BX + X^3 - 54. \tag{B.1}$$

Starting from $C_0(X) = X$ and applying the signed monic Hecke recursion of definition 9.14 and proposition 9.17 [32, 44, 45] gives

$$\begin{aligned} C_1(X) &= X^3 - 54, \\ C_2(X) &= X^6 - 1458X^3 + 39366, \\ C_3(X) &= X^{12} - 2003508X^9 + 31313466X^6 \\ &\qquad + 1230187500X^3 - 16607531250. \end{aligned}$$

The all-subgroup equations are $H_1 = C_1$, $H_2 = XC_2$, and $H_3 = C_1 C_3$. Their reductions are a^3, a^7, and a^{15}, in agreement with $N_{2,r} = 2^{r+1} - 1$.

The mod-4 carry gives $C_j \equiv 2$ modulo $(4, C_i)$ for $i < j$. It follows that

$$v_2 \operatorname{Res}(H_{r-1}, H_r) = 2^{r+1} - r - 2$$

in the first cases and, by the argument of section 9.5, at every depth. These explicit polynomials are not used in the proof; they provide a direct check of the normalization of the source and target parameters.

Statement on AI use

We used AI tools primarily to help draft proofs of results we expected to be straightforward, and occasionally to explore one or two generalizations in parallel with arguments worked out by hand. They did not guide the overall research direction, although they sometimes suggested individual problems. We also used them for informal critical review, grammatical editing, and generating Magma/Sage scripts. The authors take responsibility for the final mathematical content.

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